Department Magazine

Department of Electronics and Communication Engineering

Kalasalingam University

April 2013

Volume 4 Issue 8

ElectroComm

FIND YOURSELF...

From the Chief Editor's Desk

Dear Readers,

Being the Head of the Department and also Editor-in-Chief of this magazine, I feel happy to pen this message by presenting the *EIGHTH ISSUE* of our Department magazine *"ELECTROCOMM"*. It is the continuation of series of Department magazine, which is a mirror of the academic and creative activities of the Department and the Students. A Magazine is a good medium which encourages and inspires the students to develop their creative thinkings and to express their ways and views on various subjects & problems.This time the responsibility is bestowed upon us. Initially, it seemed like a piece of cake to publish a magazine but eventually we realized that *"ELECTROCOMM"* was like bringing down the moon on earth. Collection of articles was another hectic task. Thus bringing out a magazine is a very healthy practice to provide a joy of creation. Besides other things, the achievements of our staff's & students in different fields are also reflected in the magazine. Hence the publication of the magazine has got its own significance. I take this opportunity to congratulate the **Editor & the *Magazine Team members*** and all the contributors of this magazine for bringing out this 8^{th} Issue in time.

My best wishes for the success of this endeavor !

Dr.S.Durairaj,
Editor – in – Chief
ELECTROCOMM
(Department Magazine)

KALASALINGAM UNIVERSITY

DEPARTMENT OF
ELECTRONICS & COMMUNICATION
ENGINEERING

ELECTROCOMM

Issue- 8

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Ms. S.Naseema Palkees IIECE

"Team work divides the task & multiplies the success"

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MEMORABLE MOMENTS OF MAGAZINE

ELECTROCOMM: 1st Issue on 30th October'09

The first issue of the magazine was released by Mr. K. Venkataramanan, ITS, Deputy Manager, BSNL, Virudhunagar Telecom and it was received by Prof.S.Raju, NSS Coordinator (Retd.), Madurai Kamaraj University.

ELECTROCOMM: 2nd Issue on 22nd April'10

The second issue of the magazine was released by Dr.C.Sivapragasam, Prof. & Head of Civil Department, KLU and it was received by Dr.P.Subbaraj, Principal, Theni Kamavar Sangam Institute of Technology, Theni and Mr.Sudhakar Gummadi, Project Director, Network Security Group, TIFAC CORE, KLU

ELECTROCOMM: 3rd Issue on 23rd September'10

The third issue of the magazine was released by Mr.S.Annamalai, News Editor, THE HINDU, Madurai and it was received by Mr.A.Ashok, Serial Entertainer, SUN TV.

ELECTROCOMM: 4th Issue on 8th February'11

The fourth issue of the magazine was released by Dr.Mrs.R.Sukanesh, Professor, Department of ECE Thiagarajar College of Engineering and was received by Dr. P. Venkumar, Professor, Dept of Mechanical Engineering, KLU.

ELECTROCOMM: 5th Issue on 8th September'11

The fifth issue of the magazine was released by Dr. S.R.Srikumar, Principal, Kalasalingam Institute of Technology and it was received by Mr.Kovai Magesh, Mr.David Vadivel, Mr. Nagesh Chellakkannu, SUN TV, Artiste.

ELECTROCOMM: 6th Issue on 5th April'12

The sixth issue of the magazine was released by Mrs. Madhumitha, Writer and it was received by Dr. Nampoothari, Professor, Dept of Civil Engineering, KLU.

ELECTROCOMM: 7th Issue on 1st October'12

The seventh issue of the magazine was released by Prof. Dr. S. Jeyapragasam, Director, and International Gandhian Institute for Non-Violence and Peace(IGINP),CESCI & Rajarajan Institute of Science (RISE) & received by Dr.S.Kannan,Dean-Admissions & HOD/EEE, Kalasalingam University. Articles about lifetime achiever and Technical & Non-Technical Quiz'es are included in this issue

ACADEMIC ACHIEVERS

Final Year

Mr.Sajag Satayu Prasad Singh
ECE-D
CGPA : 9.48

Mr.Chandan Kumar
ECE-B
CGPA : 9.36

Ms.Keerthana Priya
ECE-C
CGPA : 9.18

Prefinal Year

Ms.Pusarla Swarna Madhuri
ECE-B
CGPA : 9.36

Mr.Mritunjay Kumar
ECE-B
CGPA : 9.05

Ms.V.Gomathi
ECE-A
CGPA : 9

Second Year

Ms.K.U.Roshna
ECE-C
CGPA : 9.4

Ms.T.Dhuwarahga Shri
ECE-A
CGPA : 9.0

Ms.Simmijaiswal
ECE-C
CGPA : 8.9

STUDENT'S FOOTPRINTS ON THE SANDS OF TIME

Mr. J.Vigneshwar of Final Year ECE-E sec has participated and presented a paper entitled **"FUTURE POLLING SYSTEM USING CLOUD COMUPTING"** on **QUEST O TRONICS** and secured 3rd prize at **Amruthahini College of Engineering ,Sangamner, Maharastra** held during 30th & 31st January 2013.

Ms.N.T.Theejhasri of IInd year,ECEdid an extraordinary job in collecting stamps. She has collected more than **270 stamps of 30 countries.**

Mr.SOMASUNDARAM (final year ECE-E has received best NSS student award from **Dr.K.R .MURALI held during "ENIAC-12" At April-2012**

R.BADRINATH of IInd year,ECE ECE-E has won SECOND PLACE in state level inter collegiate basket ball tournament FOR MEN organized by MARY MATHA COLLEGE, PERIYAKULAM.
(13th-15th FEB 2013)

PAPER PRESENTATION

NAME OF THE STUDENT	TITLE OF THE PAPER	PRESENTED AT
M.VINOTH KUMAR R. SETHU SATHISH	INSTRUMENTATION ROBOTICS	GENESIS 2012 HCL CAREER DEVELOPMENT CENTRE COIMBATORE. (AUGUST 2012)
J.SHABARISH SANJU R.SIBI	NANO ROBOTICS	
M.VISHNU VARDHAN S.GAUTHAM	4G TECHNOLOGY	
S.V. VIGNESH M.VIGNESH	HELIOTROPIC GADGET	
K.RAGHAVAN R.POUN RAJA	ANDROID	
B.PRASANTH P.PRASANTH	COLLEGE ATTENDANCE MANAGEMENT SYSTEM	
R.DIWAKER G.JOTHI PRAKASH	MYMO CARD (TOWARDS CURRENCY LESS ECONOMY)	
T.M. RAJKUMAR M.RAJASEKARAN	NEAR FIELD COMMUNICATION	
J.JOHSAPATH V.KARTHICK	FREQUENCY MODULATED IN PORTABLE DEVICES (FREMOPOD)	
M.VIGNESHWARAN R.VIGNESH	CANCELLATION OF CELLULAR INTERFERENCE	
S.SOWMIYA M.SRIRAM PRIYA	EASY FOR ECE .COM	

SPORTS

NAME OF THE STUDENT	EVENT	CATEGORY	POSITION	ORGANISED BY
R.ANANDHA MAHARAJAN	WRESTLING	STATE LEVEL 73-81 kg FREE STYLE JUNIOR	SILVER MEDALIST	TAMIL NADU BELT WRESTLING ASSOCIATIO N. SRI RAGHAVENDRA POLYTECHNIC COLLEGE NAMAKKAL. (1st-3rd FEB 2013)
S.KATHIRAVAN	WRESTLING	STATE LEVEL 50-55 kg CLASSICAL STYLE CADETS	BRONZE MEDALIST	
R.MUKUNTHAN	WRESTLING	STATE LEVEL 55-60 kg FREE STYLE JUNIOR	BRONZE MEDALIST	
B.MOHAMED IBRAHIM GANI	WRESTLING	STATE LEVEL 66-73 kg FREE STYLE JUNIOR	BRONZE MEDALIST	

STATE LEVEL WORKSHOPS

NAME OF THE STUDENT	TITLE	VENUE
J.JOSHAPATH JOHNSON SUNDAR SINGH	SCILAB WORKSHOP	KALASALINGAM UNIVERSITY (4th OCT 2012)
M.RAJ NIVASH	NATIONAL SERVICE SCHEME (THEME: VIBRANT YOUTH DEVELOPMENT)	THIRUVALLUVAR COLLEGE PAPANASAM TIRUNELVELI. (28th TO 30th SEP 2012)
S.SAKUNTHALA DEVI @ PRIYA		

STAFF ACHIEVEMENTS

- **Dr. S. DURAIRAJ Sr. Prof & Head Department of ECE delivered Guest lectures on**
  - **FACTS & its Applications** in the AICTE sponsored Technical Education Quality Improvement Programme (TEQIP) duly organized by A.C. College of Technology Karaikudi during 6th Nov 2012
  - **Industrial automation** in the DRDO sponsored seminar during 9th Nov 2012 at Bharat Niketan Engineering College.

- **Various positions held by Dr.S.Durairaj**
  - **Nominated as Board of Management member (BOM) of our university**
  - **Elected as NBA team expert member from National Board of Accreditation(NBA)**
  - Member of Board of Studies in **Karpagam Univerity Sri Chandrasekharendra SaraswathiViswa Mahavidyalaya Dr.M.G.R. University SRM University PSNCET**
  - Conference Adivisory /Technical Committee in Sri Chandrasekharendra SaraswathiViswa MahavidyalayaDr.M.G.R. UniversitySRM UniversityNEC.
  - **Doctoral Committee Members : for Research Scholars in Anna University Madurai.**
  - Chaired the technical sessions of National and International Conferences in **KSRCET NEC PSNACET Velammal CETMEPCO Schlenk College of Engg. & Tech.**

- **Dr. S. DURAIRAJ has publications of**
  - **National conference : 2**
  - **International conference : 2**
  - **Journal : 2**

- **Prof.V.K. MADAN – "TELEHEALTH" at the seminar on Advanced Technology and International collaborations held on FEB 12th 2013 at Krishnankoil.**

- **Chitradeep Majumdar S.N.Merchant V.K.MADAN "ENERGY EFFICIENT COGNITIVE RADIO AND MIMO BASED SENSOR NETWORKS" proceedings of International Conference on Instrumentation Communication Control and Automation (ICICCA-2013) Jan 3-5 Krishnankoil.**

- **H.S.Kamath V.K.MADAN "STATISTICAL PROPERTIES OF AMPLIFY AND FORWARD RELAY FADING CHANNELS" proceedings of International Conference on Instrumentation Communication Control and Automation (ICICCA-2013) Jan 3-5 Krishnankoil.**

- **Mr.B.Perumal received BEST MOTIVATOR AWARD from Kavingar & Actor "SNEGAN" at Meenakshi Mission Hospital Madurai.**

- Mr JENYFAL SAMPSON & Mr .S.P.VELMURUGAN - Participated in "ORIENTATION TRAINING" conducted by "EMPANELLED TRAINING INSTITUTION – NATIONAL SERVICE SCHEME" from 14.02.2013 to 20.02.2013.

The Virtues of a Teacher

- ✓ To be a teacher is to have a fair share of selflessness and service.
- ✓ To be a teacher is to reflect patience and purity in his conduct.
- ✓ To be a teacher is to be heightened by the zeal of life.
- ✓ To be a teacher is to be flavored with courtesy and kindness.
- ✓ To be a teacher is to be tempered by impartiality and integrity.
- ✓ To be a teacher is to be sustained by awareness and alertness.
- ✓ To be a teacher is to be enriched by love and sympathy.

DEPARTMENTAL ACTIVITIES

The following are the notable activities in the ECE department from October 2012 to 20th April 2013.

DATE	NAME OF THE EVENT	Organized By	RESOURCE PERSONS
Oct 1st 2012	**Wii – Pressenza & Magazine Release (ELECROCOMM-7th Issue)**	Wireless Mavericks club of ECE &ECE Association	Dr. S. Jeya Pragasam, Director, International Gandhian Institute for Nonviolence and Peace (IGINP), CESCI & Rajarajan Institute of Science (RISE).
Nov 2nd & 3rd 2012	**International Conference on Electronics, Communication &Information systems** ICECT12	Department of ECE	•Prof. **Liyanage Chandratilak De Silva,** Associate Professor ,University of Brunei, Brunei Darussalam. •Prof. Arup Kumar Gogoi , Professor,Department of Electronics and Electrical Engineering,IIT, Guwahati •Prof.V.Vaidehi, Professor and Head, Department of Information Technology MIT, Anna University, Chennai •Dr.C.Tharani, Professor,Dept. of ECE,B.S.Abdur Rahman University,Chennai.
Feb 3rd 2013	Faculty development program on **"Life skill management"**	Dean - CLT	•Prof. Mayil Murugan •Prof. Helan Christina •Shri.X.P.White from Madurai Institute of Social Science.
Feb 25th 2013	**KLU – NATURA'13 (Student Fest)**	Wireless Mavericks Club & IETE	Contest on the theme of Natural Resources.
March 5th 2013	Career Guidance Program	ECE Association	IMS Learning Services Madurai
	Opportunities at Abroad		Mr. Vidyasahar Research Assistant Germany.
March 7th 2013	One day National Seminar on **"E-Waste Management"**	Organizing Committee	• Prof. S. Ramaswamy Gandhigram Rural University • Prof. M.Ravichandran Bharathidasan University
April 12th	National Level Technical Symposium **"ENIAC 2013**	Organizing Committee	• Department of ECE
April 12th 2013	Magazine Release **"Electrocomm Issue 8"**	Magazine Committee	
Jan 10 2013 Feb 2 2013 March 2 2013 April 3 2013	**External Counseling**	• Sri X.P.White External Counselor cum Writer Madurai.	

Techie Articles

THE WEIRDEST INVENTIONS OF CES 2013

2013 International CES

The CES (Consumer Electronics Show) happened at Las Vegas @ 8th to 13th January. There were many products displayed by several companies including Samsung, Sony, Philips, HTC, etc. Here are some gadgets which were very weird.

Armortech:

The new Armortech film is made from a Poly-Ethylene and it claims to boost the shatter resistance of any LCD & LED by 220%. It keeps your delicate Smartphone display from cracking. To demonstrate, they had a hammer continuously pound on the phone's display.

Power bright:

This power inverter turns your car's 12V cigarette lighter into an AC wall outlet as well as a USB port charger.

Party Animals:

These dancing stuffed animal dolls double as music speakers as they move their torsos, heads, and arms, making them the cutest Bluetooth speakers at CES.

Hype Bluetooth Mobile Power Prism:

The Power Prism connects to your Smartphone via Bluetooth, turning it into a traditional cordless/speaker phone.

Dri Suit:

Dri Suit not only makes the Iphone waterproof, but also allows for use when submerged. The device creates a protected channel for the headphone jack for listening to music underwater.

"Live as if you were to die tomorrow. Lear as if you were to live forever."
-Mahatma Gandhi

Genius Ring Presenter:

This little gadget is worn like a ring and allows you to control your desktop cursor by waving it through the air.

By,
T.M.Raj Kumar
Second Yr Ece-B

POWER GRID : GUJARAT SETS AN EXAMPLE

AHMEDABAD: A power surplus state with near 24-hour electricity supply not just in cities like Ahmedabad and Vadodara but in all the 18,000 villages. Now, the Gujarat government plans to further sharply increase power generation from 13,500 MW now to 18,000 MW by the end of the current year.

The Narendra Modi government was able to ensure almost 24 hour electricity supply, especially in villages, by implementing the Jyoti Gram project. Even the Government of India has accepted this as a flagship scheme for the 12th Five-Year plan (2012-17) for supplying round-the-clock, high-quality, three-phase power to all villages.

Commissioned in 2006, Jyoti Gram provides for a separate electric feeder for domestic use and a limited agricultural supply of nearly eight hours a day, continuous and of constant voltage. A recently released Planning Commission document, "Faster, Sustainable and More Inclusive Growth : An Approach to the 12th Five Year Plan", says "the separation of agricultural feeders" in the country will enable villages to get "24 X 7 three-phased power for domestic uses, schools, hospitals and village industries".

As for the farm pumpsets, which require more power, they can obtain "eight hours or more of quality power on a pre-announced schedule." The document underlines, "The programme of feeder separation has to be carried through across the country. Gujarat has achieved very good results by combining feeder separation with an extensive watershed programme for groundwater recharge. Punjab, Karnataka, Andhra Pradesh, Maharashtra and Madhya Pradesh have also moved forward in this direction. Feeder separation needs to be extended to all states, especially where groundwater is extensively used."

The Gujarat government spent Rs 1,200 crore to implement Jyoti Gram by separating 12,000 agricultural feeders from domestic feeders. It brought down transmission and distribution losses from 35 per cent five years ago to 15-19 per cent this year.

Already a power surplus state, Gujarat sold 5,105.43 million units (MUs) to other states last year earning a profit of Rs 1,888.53crore. Last year, the state had sold 5,105 million units to states like Rajasthan, Haryana, Punjab, Delhi and Maharashtra. This was approximately seven per cent of total power produced in the state — 68,710 MUs. According to minister of state for power, Saurabh Patel the government sold power at Rs 8.51 per unit to Rajasthan, at Rs 7.70 per unit to Maharashtra and 9.52 per unit to Delhi. By selling these surplus power, the government was giving Rs 3,000 crore as subsidy to farmers.

With new plants planned to come up, the situation would further improve. And, Gujarat is not just planning to have more imported coal and gas based power plants, but is also negotiating with the Government of India for a second ultra mega power plant (UMPP). There has been no looking back since 2004 when the state successfully unbundled the loss-making Gujarat Electricity Board (GEB) into smaller power utilities. Smaller set-ups improved efficiency - cutting T&D losses and better plant load factor - helping the firms to make profits.

PRIZE WINNER

By,
C.Nanthini
Second Yr Ece-B

"Yesterday is history, tomorrow is a mystery, today is a gift of God, which is why we call it the present." ? Bil Keane

B&O ADVANCED SOUND SYSTEM

Top-notch factory audio for the Audi A8 and S8.

The 8-inch, rear deck, subwoofer - taken at the New York Auto Show

With the debut of the Bang &Olufsen Advanced Sound System in the Audi A8 and S8, the Danish audio-design icon and German luxury carmaker have raised the performance bar - and price - of premium OEM sound systems. B&O's new benchmark system was unveiled in January at the North American International Auto Show in Detroit in the sleek '07 Audi S8 - as a mere $6,700 option on the $100K super luxury-performance sedan. (The system will be available in the '07 Audi A8 this summer and in the S7 in the fall.)

This costly add-on gets you 14 speakers powered by over 1,000 watts. True to B&O's heritage, the system looks stunning, with stylish aluminium speaker grilles integrated into the car's doors and rear deck. But the real eye (and ear) candy is a pair of Acoustic Lens tweeters that rise out of the corners of the dash when the system is turned on (and retract when it's switched off).

The motorized tweeters are complemented by a 70-mm center channel in the dash, a 90-mm midrange and 133-mm midbass driver in each front door, a 25-mm tweeter and 140-mm midbass in each rear door, and two 90-mm full-range drivers and a 200-mm subwoofer in the rear deck. B&O's ICE power five-channel amplifier drives the sub with 250 watts and the four door-mounted midbass speakers with 125 watts each; a MOST DSP amp powers the rest of the speakers with 28 watts each.

Of course, the $6,700 question is: How does it sound? In a word: awesome. During a demo at the Detroit Auto Show, I listened to a CD of Danish female jazz vocalist JetteTorp. Tonal detail was excellent, the soundstage extended beyond the boundaries of the interior, and imaging was pinpoint accurate. The bass response at the driver's seat was deep, tight, and powerful, with no smearing to the rear, despite the sub being in the back deck.

The system rocks, too. I cranked up AC/DC's pulverizing "Thunderstruck" as much as my ears could stand, and the system continued to play cleanly, smoothly, and very loudly.

The Audi B&O stock system ranks with some of the best aftermarket systems I've ever heard.

By,
M.Sivaraman
Second Yr Eco-C

MYMO CARD TOWARDS CURRENCY LESS ECONOMY

Introduction:

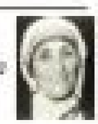
The proposed MYMO (My Money) card is a new type of bio-metric chip based card used to transfer money between two parties without physical handling of paper currency. In due course the improvement in MYMO card technology will certainly lead to currency less economy. It would have both receipts and payments option of money value.

Uses: MYMO card is very useful for transferring money value between two parties.

The transfer may be from

- Individual to individual
- Individual to bank
- Individual to government
- Individual to shopping points
- Individual to any organization offering goods and services and
  Vice- versa.

"*Not all of us can do great things. But we can do small things with great love.*" ? Mother Teresa



Requirements:

- A special and secured network system developed by central bank of a nation to facilitate MYMO card operation.
- Hardware and software installation at banks to connect the MYMO cards and enable the transaction.
- A secured issue system for MYMO card through banks.
- A bio-metric chip based card for effecting payment of money value.
- A card reader instrument for effecting receipts of money value.

Working Of Mymo Card:

Illustration: Payment of Rs.10, 000 by Mr. A to Mr. B.

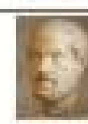
- Mr. A inserts his MYMO card to the card reader of Mr. B with his finger on it.
- The card reader reads the finger print and sends the data to the central system.
- On recognizing the finger print, the central system would send the data the concerned bank for further processing.
- After the verification of the data by bank the reader prompts for entering the payment details.
- Mr. A enters the amount using the keyboard on the reader.
- After the processing by bank, Mr. A's card is debited by Rs.10, 000 and B's card is credited with Rs.10, 000.
- The confirmation message about the transfer of money value is displayed on the screen.
- Mr. A can remove the card and finish the transaction.

DIFFERENCES BETWEEN CREDIT / DEBIT CARD AND MYMO CARD:

CREDIT / DEBIT CARD	MYMO CARD
➢ Anyone can use the card if he knows the password.	➢ Only the original owner of the card can use since it is based on the bio-metric technology.
➢ Handling of paper currency is involved while withdrawing cash.	➢ It eliminates paper currency handling for both receipts and payments.
➢ Comparatively less secured.	➢ Comparatively more secured.
➢ It requires very big and costly hardware like ATM machines.	➢ It does not require such huge investment in hardware.

Benefits:

- The government's very critical and tedious job of printing and managing currency is reduced considerably.
- The physical handling of currency at all levels is minimized.
- The need for safety locker and safe vault at home to save and preserve the money is considerably reduced.
- The serious socio-economic evils such as black money, corruption and bribing can be reduced to larger extent, since all money transactions are transparent and accounted for by a centralized system.
- The banks operation in physical handling of currency is minimized.



Commercial Importance:

The proposed MYMO card is commercially very viable project for hardware and software manufacturing and marketing companies. It opens new vistas in marketing by providing new consumer segment.

Working Of Bio-metric Chip:

In this MYMO card we use biometric chip based sensor for scanning our thumb. When we insert our card to the card reader, we must insert using our thumb and the thumb must be placed in the sensor panel. When the card is inserted power activation from card reader to card takes place and thumb gets scanned in the sensor panel. And finally information from the sensor panel is passed to the MYMO card reader and transaction takes place.

Advantages Of Mymo Card:

- More secured – Only the owner of the card can use it.
- Leads to currency less economy.
- Safer transaction of money.

Future Economy:

If MYMO card is implemented by our government to the society activities, it will be a great boom to our Indian society. Since MYMO card helps in transaction of money from one point to another point without any form of paper currency or receipts. There is not a need of money in this transaction. So it is currency less. And this will lead to CURRENCY LESS ECONOMY, the economy which needs for the Indian society. Consider an economy without money; it will be a great innovation towards the world. By introducing this economy, BLACK MONEY gets vanishes, humanity gets developed, and nation improves on more developmental aspects.

Conclusion:

The concept of MYMO card is very useful to any country which wants to move towards the currency less economy. The farther developments and advancements in the working of MYMO card technology will certainly present the world a new innovation in the secured money transfer.

By,
J.Joshapath
Second Yr, Ece-A

FREMOPOD

Introduction:

"FREMOPOD"-means or refers to ***"FREQUENCY MODULATION IN PORTABLE DEVICE"***. Our paper is to give you an idea about tapping current from radio waves in a more economical manner by

modifying the circuit and working of a normal radio receiver. We are attracted about this idea by considering our nation's economy and rural development. We named this paper as FREMOPOD as we are going to work with modulation in frequency to tap current and as this device is portable we named it as above.

Guided By Tesla:

As we know transmitting current without wires have been always a dream for scientists because, even it is possible it costs much more than transmission of current through wires. Even great scientists like NICHOLAS TESLA worked on this subject. Tesla is considered as the first person to seed this idea in this world. He also tried his best to transmit current without wires by using TESLA TOWER (constructed by himself for wireless transmission of current in New York). But he was unsuccessful as this costs many times more than wired transmission.

Working Of A Radio Transmitter:

In a typical transmitter high frequency radio waves are sampled with same frequency current waves. These current waves act as carrier waves. These high frequency current signals are transmitted through high resonant antenna terminals. These current signals dissipate into air in the form of electromagnetic waves. While at the point of transmission the strength of the electromagnetic waves is higher, but due to natural, human and other radio devices interruption it reaches the receiver as a weak wave.

Working Of A Radio Receiver:

Similar to the working of transmitter, receiver also receives the radio waves with the help of an antenna. A suitable or frequency matching to the radio wave to be received is created in the antenna using a high frequency oscillator. When matching electromagnetic waves strike the antenna, it gets absorbed by the antenna. Electromagnetic waves again convert themselves into current signals i.e., carrier wave.

Detector Or Demodulator:

Detector is the word used in the period of origin of morsecodes. And now it has survived in the form of the word "Demodulator ". Carrier waves normally have both phases, but for a radio to work only we need a single phase. A Diode can be called as a simple demodulator as it allows only waves with a single phase to pass through it. But modern demodulator uses so many techniques to separate required radio waves from the unwanted noise (current signals).

Twist In The Story:

But according to our idea, if we remove the demodulator from the radio circuit the wave remains in the form of current signal that is nothing but low strength current. And if we use step-up transformers we can dramatically increase the voltage of the received current signal and by amplifying it we can get a continuous range of current.

Need To Do This?

- Consider that you are travelling in a bus suddenly your mobile rings and blinks in the name of your boss and goes off as the battery level runs out .... What may be the possible way to recharge it again? We are sure that you need our FREMOPOD there as it is portable power generator exactly power tapper.

In India still there are rural and forest areas where people can't see a single electrical wire in the sky running from the electricity board, but I am sure that

- there is no place in this world that has escaped from radio waves. At present we can say "wherever there is air, there is a radio wave ". So at least we can light a single bulb to show their night with light.
- And as a major point it is freeee.... We don't have to setup TESLA TOWER to transmit current with more cost. We are just going to utilize the already existing radio waves. It's more economical to our country.
- We can get an uninterrupted power supply both day and night.

Conclusion:

However this idea and not a practical one. And we are eager to implement this in a practical manner .but we have to get a seal that we are worth to do this, and that is vested in the hands of yours. Hope we did our part better.

By,

R.Diwaker

Second Yr, Ece-A

"The unexamined life is no worth living." - Socrates

BUILD A PORTABLE PASSWORD GENERATOR

PINs, passwords, secure entry systems; just about everything you access these days needs some form of passive authentication code. Heck, even that Wi-Fi router in your basement uses a mega63-character password for enabling secure access. Before you know it, you'll even need a password just to go to bed at night.

More often than not, these security code systems require you to enter a lengthy 8- or more digit alphanumeric sequence. And creating powerful, unique, protective passwords can be a tough challenge for most people. What you need is some sort of portable, automatic, press-the-button-and-read-the-password device. Pass Key is an attempt at satisfying this requirement.

Built from a modified AVR development board manufactured by Olimex, Pass Key is a sweet little 5V "candy bar" style computer that can generate 16-digit alphanumeric passwords as fast as you can press a button. Yes, the original AVR-MT dev board requires 10-14V, but with a little circuitry magic, this power demand can be sliced to a scant 5V without losing any of the display, buzzer, LED, button or computational capabilities of the original dev board.

Just like the more popular assembly- and C-based programming environments, BASCOM-AVR generates a sweet, tight HEX code that can be independently uploaded with any USB AVR programmer or directly burned onto an AVR with Atmel's STK500 and a serial COM port.

NOTE: If you own an AVRISP mkII USB programmer, you can still use BASCOM-AVR for developing your code. Rather than using BASCOM-AVR for programming your microcontroller, however, you will compile a HEX code for your BASIC program, and then burn the HEX code into the microcontroller with Atmel's Studio 4 IDE.

We've included both the Pass Key BASCOM-AVR code, as well as a demo BASCOM-AVR program for the stock AVR-MT dev board. Both of these programs can be downloaded, compiled within the BASCOM-AVR IDE, and programmed on the ATtiny2313 microcontroller on your Olimex AVR-MT dev board.

- Pass Key Program
- AVR-MT Demo Program
- **Time:** 2hours
  Cost: $43.01
  Difficulty: Moderate

Parts

- Olimex AVR-MT development board (Spark Fun; $38.95)
- Atmel ATtiny2313 (Spark Fun; $2.88)
- Amp 2-pin header (Mouser #571-111237232; $0.28)
- Amp 2-pin housing (Mouser #571-111237222; $0.14)
- (2) Amp contacts (Mouser #571-11237212; $0.26)
- Leather case (All Electronics #CSE-29; $0.50)
- (2) Short lengths of scrap wire (salvaged; FREE

Steps:

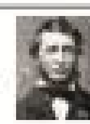
1. Convert the AVR-MT dev board to 5V power. Carefully remove the following components from the dev board: D1, C3, C1, VR, and PWR.

2. Add two jumpers. Cut the short lengths of scrap wire to approximately 1/2-inch long, strip off some insulation from each end, and bend each wire into a "U" shape. Carefully solder the first jumper between the two pins of the previously removed component: D1. Next, solder the second jumper between the input and output pins of the previously removed component: VR. These pins are the two outside pins of the three-pin VR.

3. Add the header. Solder the header into pins 2 & 3 of the previously removed component: PWR. Pins 2 and 3 are the right pin and the furthest back pin, respectively. Pin 2 is GND and pin 3 is the +5V (+) pin. Remember this Amp header is a polarized receptacle. Keep this orientation in mind for properly mating this header with the power plug from your 5V power supply.

NOTE: The Amp housing and contacts are used for mating your 5V power supply with the dev board's new power receptacle.

4. Insert the ATtiny2313 microcontroller. Press the microcontroller into the 20-pin IC socket on the AVR-MT dev board. Plug your 5V power supply into the dev board.



Inside the Passkey: The Olimex AVR board shown with the necessary components removed and the jumpers and header installed.

5. Download the BASCOM-AVR MVR-MT Demo program. Download the Demo program, create a HEX file with BASCOM-AVR, and burn this file onto the AVR-MT board with Atmel Studio 4 and your AVR programmer. Remember to turn on the 5V power supply prior to connecting the AVR programmer.

6. Run the Demo program. Ensure that the Demo program properly lights the LED, activates the LCD, and plays "Here Come the Saints" on the AVR-MT buzzer. This step will verify that your 5V power conversion is functioning correctly.

7. Load the Pass Key code. Once the Demo program checks out, download the Pass Key program, create a HEX file in BASCOM-AVR, and burn this file on the AVR-MT dev board.

8. Generate a password. Following the opening credits, Pass Key will generate a 16-character alphanumeric password for every press of button #4. If you feel snoopy eyes scoping your Pass Key, you can toggle the display on/off with button #1. Additionally, there are comments in the Pass Key code for enabling other buttons on the dev board and adding your own code for saving your generated passwords.

NOTE: The random numbers in BASCOM-AVR are generated by the compiler. Therefore, the "random" sequence will be the same every time the Pass Key program is executed. If you don't like this rote repetition, you can modify the program like this:

- Dimension a different random number seed value
- Change the length of the generated password
- Remove/exclude the use of special characters (e.g., #, ~, etc.) in the password.

If a 16-character password is not to your liking, then modify the code for enabling different password lengths via different dev board button presses. Now slip Pass Key inside its leather case and head smartly off to any security encrypted destination. By using Pass Key your password prowess should be much more robust which should help you get a better night's sleep. Good night, Pass Key.

By,
R.Muthu Selvi
Pre-Final Yr Ece-A

SMARTPHONE APPLICATION DEVELOPMENT

What Is a Mobile Phone?

A mobile phone is more frequently called a cellular phone or cell phone. These communication devices connect to a wireless communications network through radio waves or satellite transmissions. Most mobile phones provide voice communications, Short Message Service (SMS), Multimedia Message Service (MMS), and newer phones may also provide Internet services such as Web browsing, instant messaging capabilities and e-mail.

What Is a Smartphone?

A Smartphone is considered to be the combination of the traditional PDA and cellular phone, with a bigger focus on the cellular phone part. These handheld devices integrate mobile phone capabilities with the more common features of a handheld computer or PDA. Smartphone's allow users to store information, e-mail, and install programs, along with using a mobile phone in one device. A Smartphone's features are usually more oriented towards mobile phone options than the PDA-like features. There is no industry standard for what defines a Smartphone, so any mobile device that has more than basic cell phone capabilities can actually be filed under the Smartphone category of devices. Smart phones and feature phones may be thought of as handheld computers integrated within a mobile telephone, but while most feature phones are able to run applications based on platforms such as Java ME, RIM,

Android, simians, Wince. In a nutshell, a Smartphone is a device that lets you make telephone calls, but also adds in features that you might find on a personal digital assistant or a computer--such as the ability to send and receive e-mail and edit Office documents, play media and web Access, instant messaging services. Smartphone usually allows the user to install and run more advanced applications. Smart phones run complete operating system software providing a platform for application developers.

Operating System:

In general, a Smartphone will be based on an operating system that allows it to run productivity applications. BlackBerry smart phones run the BlackBerry OS, while other devices run the Palm OS or Windows Mobile. There are Smartphone OS that are pared-down versions of desktop Linux, too. Windows CE Pocket PC OS would be offered as "Microsoft Windows Powered Smartphone Microsoft

"Honesty is the first chapter of the book wisdom." — Thomas Jefferson

originally defined its Windows Smartphone products as lacking a touch screen and offering a lower screen resolution compared to its sibling Pocket PC devices.

RIM released the first BlackBerry which was the first Smartphone optimized for wireless email use and had achieved a total customer base of 32 million subscribers.Nokia launched the Nokia Smartphone Series which integrated a wide range of features into a consumer-oriented Smartphone: GPS, megapixel camera with autofocus and LED flash, 3G and Wi-Fi connectivity and TV-out. In the next few years these features would become standard on high-end smart phones. Android, a cross platform OS for smart phones was released in 2008. Android is an open source platform backed by Google, along with major hardware.

A Smartphone is a mobile phone built on a mobile operating system, with more advanced computing capability and connectivity than a feature. The first smart phones combined the functions of a personal digital assistant (PDA) with a mobile phone. Later models added the functionality of portable media players, low-end compact digital cameras, pocket video cameras, and GPS navigation units to form one multi-use device. Many modern smart phones also include high-resolution touch screens and web browsers that display standard web pages as well as mobile-optimized sites. High-speed data access is provided by Wi-Fi and mobile broadband. In recent years, the rapid development of mobile app markets and of mobile commerce have been drivers of Smartphone adoption.

The mobile operating systems (OS) used by modern smart phones include Google's Android, Apple's iOS, Nokia's Symbian, RIM'sBlackBerryOS, Samsung's Bada, Microsoft's Windows Phone, Hewlett-Packard's webs, and embedded Linux distributions such ashamed and MeeGo. Such operating systems can be installed on many different phone models, and typically each device can receive multiple OS software updates over its lifetime. A few other upcoming operating systems are Mozilla's Firefox OS and Canonical Ltd.Symbian is a mobile operating system designed for smart phones originally developed by Symbian Ltd.but currently maintained by Accenture. The Symbian platform is the successor to Symbian OS and Nokia Series 60. The latest version, Symbian^3, was officially released in Q4 2010 and first used in the Nokia N8.

By,
S.Prabha Devi
Pre-Final Yr Ece-B

NASA'S ROBOTIC REFUELING DEMO SET TO JUMPSTART EXPANDED CAPABILITIES IN SPACE

In mid-January, NASA will take the next step in advancing robotic satellite-servicing technologies as it tests the Robotic Refueling Mission.

The Robotic Refueling Mission, or RRM, investigation (center, on platform) uses the International Space Station's Canadarm2 and the Canadian Dextre robot (right) to demonstrate satellite-servicing tasks.

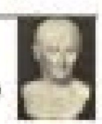
During five days of operations, controllers from NASA and the Canadian Space Agency will use the space station's remotely operated Special Purpose Dexterous Manipulator, or Dextre, robot to simulate robotic refueling in space. Operating a space-based robotic arm from the ground is a feat on its own, but NASA will do more than just robotics work as controllers remotely strip wires, unscrew caps and transfer simulated fuel. The team also will demonstrate tools, technologies and techniques that could one day make satellites in space greener, more robust and more capable of delivering essential services to people on Earth.

Why Fix or Refuel a Satellite?

"Every satellite has a lifespan and eventual retirement date, determined by the reliability of its components and how much fuel it can carry," explains Benjamin Reed, deputy project manager of NASA's Satellite Servicing Capabilities Office, or SSCO. Repairing and refueling satellites already in place, Reed asserts, can be far less expensive than building and launching entirely new spacecraft, potentially saving millions, even billions of dollars and many years of work.

The RRM demonstration specifically tests what it would take to repair and refuel satellites traveling the busy space highway of geosynchronous Earth orbit, or GEO. Located about 22,000 miles above Earth, this orbital path is home to more than 400 satellites, many of which beam communications, television and weather data to customers worldwide.

By developing robotic capabilities to repair and refuel GEO satellites, NASA hopes to add precious years of functional life to satellites and expand options for operators who face unexpected emergencies, tougher economic demands and aging fleets. NASA also hopes that these new technologies will help boost the commercial satellite-servicing industry that is rapidly gaining momentum.



Besides aiding the GEO satellite community, a capability to fix and relocate "ailing" satellites also could help manage the growing orbital debris problem that threatens continued space operations, ultimately making space greener and more sustainable.

How RRM Is Making a Difference?

Built by SSCO in the span of 18 months, the washing-machine-sized RRM module contains the components, activity boards and tools to practice several of the tasks that would be performed in orbit during a real servicing mission. Launched to the space station on July 8, 2011, aboard the final mission of the Space Shuttle Program, RRM was the last payload an astronaut ever removed from a shuttle.

In 2012, RRM demonstrated dexterous robotic operations in space. Dexter's 12-foot arm and accompanying RRM tool successfully snipped two twisted wires -- each the thickness of two sheets of paper -- with only a few millimeters of clearance: a task essential to the satellite refueling process.

The RRM refueling demonstration on Jan. 14-24 will employ the Canadian-built Dextre, NASA's RRM module and four unique RRM tools to show

that space robots controlled from Earth -- hundreds or even thousands of miles below -- can transfer fuel to satellites with triple-sealed valves that were never designed to be accessed.

"The RRM operations team is very excited about the upcoming refueling demonstration," says Charlie Bacon, RRM operations manager. "Over the last two years, the team has put in more than 300 hours of preparation -- reviewing procedures, running simulations, and communicating with team members from other NASA centers and our international partners. When we finally execute the namesake task of RRM, we anticipate that our work will culminate in proving that in-orbit satellite refueling is no longer future technology -- its current technology."

Although the RRM module will never fix or refuel a satellite itself, its advanced tools and practice runs are laying the foundation for future in-orbit robotic servicing missions. Additional RRM demonstrations will continue into 2013, with a new round of servicing task boards, tools and activities slated to continue its investigations through 2015.

What's Next in Robotic Satellite Servicing?

The satellite-servicing concept that RRM is advancing is one that NASA has been developing for years. Beginning with the Solar Maximum repair mission in 1984, the servicing philosophy paved the way for five successful astronaut-based missions to upgrade and repair the Hubble Space Telescope and has been practiced more recently in spacewalks to assemble and maintain the space station.

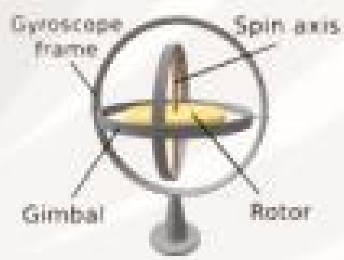
With the RRM on the space station and a robust technology development campaign being conducted on the ground, NASA is testing capabilities for a new robotic servicing frontier. Since 2009, the Satellite Servicing Capabilities Office at NASA's Goddard Space Flight Center in Greenbelt, Md., has been aggressively advancing the robotic technologies for a free-flying servicer spacecraft that could access, repair and refuel satellites in GEO. To this end, the SSCO team has been studying a conceptual servicing mission and building technologies to address uncharted territory such as autonomous rendezvous and docking, propellant transfer systems for zero gravity and specialized algorithms (computer commands) to orchestrate and synchronize satellite-servicing operations. Systems engineering review on this conceptual

mission was recently conducted with positive responses from peer experts and external participants.

By,
S.Prabha Devi
Pre-Final Yr Ece-B

GYROSCOPE

A **gyroscope** is a device for measuring or maintaining orientation, based on the principles of angular momentum. Mechanically, a gyroscope is a spinning wheel or disk in which the axle is free to assume any orientation. Although this orientation does not remain fixed, it changes in response to an external torque much less and in a different direction than it would without the large angular momentum associated with the disk's high rate of spin and moment of inertia. The device's orientation remains nearly fixed, regardless of the mounting platform's motion, because mounting the device in a gimbals minimizes external torque.

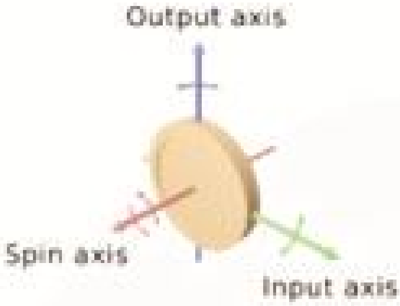


Gyroscopes based on other operating principles also exist, such as the electronic, microchip-packaged MEMS gyroscope devices found in consumer electronic devices, solid-state ring lasers, fiber optic gyroscopes, and the extremely sensitive quantum gyroscope.

Applications of gyroscopes include inertial navigation systems where magnetic compasses would not work (as in the Hubble telescope) or would not be precise enough (as in ICBMs), or for the stabilization of flying vehicles like radio-controlled helicopters or unmanned aerial vehicles. Due to their precision,

gyroscopes are also used to maintain direction in tunnel mining.

Description and diagram:



Within mechanical systems or devices, a conventional *gyroscope* is a mechanism comprising a rotor journeyed to spin about one axis, the journals of the rotor being mounted in an inner gimbals or ring; the inner gimbals is journeyed for oscillation in an outer gimbals for a total of two gimbals.

The **outer gimbals** or ring, which is the gyroscope frame, is mounted so as to pivot about an axis in its own plane determined by the support. These outer gimbals possess one degree of rotational freedom and its axis possesses none. The next **inner gimbals** is mounted in the gyroscope frame (outer gimbals) so as to pivot about an axis in its own plane that is always perpendicular to the pivotal axis of the gyroscope frame (outer gimbals). These inner gimbals have two degrees of rotational freedom.

The axle of the spinning wheel defines the spin axis. The rotor is journeyed to spin about an axis, which is always perpendicular to the axis of the inner gimbals. So the rotor possesses three degrees of rotational freedom and its axis possesses two. The wheel responds to a force applied about the input axis by a reaction force about the output axis.

The behavior of a gyroscope can be most easily appreciated by consideration of the front wheel of a bicycle. If the wheel is leaned away from the vertical so that the top of the wheel moves to the left, the forward rim of the wheel also turns to the left. In other words, rotation on one axis of the turning wheel produces rotation of the third axis.

A **gyroscope flywheel** will roll or resist about the output axis depending upon whether the output gimbals are of a free- or fixed- configuration. Examples of some free-output-gimbals devices would be the attitude reference gyroscopes used to sense or measure the pitch, roll and yaw attitude angles in a spacecraft or aircraft.

The centre of gravity of the rotor can be in a fixed position. The rotor simultaneously spins about one axis and is capable of oscillating about the two other axes, and, thus, except for its inherent resistance due to rotor spin, it is free to turn in any direction about the fixed point. Some gyroscopes have mechanical equivalents substituted for one or more of the elements. For example, the spinning rotor may be suspended in a fluid, instead of being pivotally mounted in gimbals. A control moment gyroscope (CMG) is an example of a fixed-output-gimbals device that is used on spacecraft to hold or maintain a desired attitude angle or pointing direction using the gyroscopic resistance force.

In some special cases, the outer gimbals (or its equivalent) may be omitted so that the rotor has only two degrees of freedom. In other cases, the centre of gravity of the rotor may be offset from the axis of oscillation, and, thus, the centre of gravity of the rotor and the centre of suspension of the rotor may not coincide.

History:

Essentially, a gyroscope is a top, a self-balancing spinning toy, put to instrumental use. Tops were invented in many different
civilizations, including classical Greece, Rome, India and China, and the Maori culture a thousand years later. Most of these, though using the same conservation of angular momentum as a gyro, were not utilized as instruments.

The first known use of such a top as an instrument came in 1743, when John Serson invented the "Whirling speculum" (or Serson's Speculum), a spinning top that was used as a level, to locate the horizon in foggy or misty conditions.

The instrument used more like an actual gyroscope was made by German Johann Bohnenberger, who first wrote about it in 1817. At first he called it the "Machine". Bohnenberger machine was based on a rotating massive sphere. In 1832, American Walter R. Johnson developed a similar device that was based on a rotating disk.[10] The French mathematician Pierre-Simon Laplace, working at the École Polytechnique in Paris, recommended the machine for use as a teaching aid, and thus it came to the attention of Léon Foucault. In 1852, Foucault used it in an experiment involving the rotation of the Earth. It was Foucault who gave the device its modern name, in an experiment to see (Greek *skopeein*, to see) the Earth's rotation (Greek *gyros*, circle or rotation), which was visible in the 8 to 10 minutes before friction slowed the spinning rotor.

In the 1860s, the advent of electric motors made it possible for a gyroscope to spin indefinitely; this led to the first prototype heading indicators and, quite more complicated devices, first gyrocompasses. The first functional gyrocompass was patented in 1904 by German inventor Hermann Anschütz-Kaempfe. The American Elmer Sperry followed with his own design later that year, and other nations soon realized the military importance of the invention—in an age in which naval prowess was the most significant measure of military power—and created their own gyroscope industries. The Sperry Gyroscope Company quickly expanded to provide aircraft and naval stabilizers as well, and other gyroscope developers followed suit.

In 1917, the Chandler Company of Indianapolis created the "Chandler gyroscope", a toy gyroscope with a pull string and pedestal. Chandler continued to produce the toy until the company was purchased by TEDCO inc. in 1982. The chandler toy is still produced by TEDCO today.

In the first several decades of the 20th century, other inventors attempted (unsuccessfully) to use gyroscopes as the basis for early black box navigational systems by creating a stable platform from which accurate acceleration measurements could be performed (in order to bypass the need for star sightings to calculate position). Similar principles were later employed in the development of inertial guidance systems for ballistic missiles.

During World War II, the gyroscope became the prime component for aircraft and anti-aircraft gun sights. After the war, the race to miniaturize gyroscopes for guided missiles and weapons navigation systems resulted in the development and manufacturing of so-called **midget gyroscopes** that weighed less than 3 ounces (85 g) and had a diameter of approximately 1 inch (2.5 cm). Some of these miniaturize gyroscopes could reach a speed of 24,000 revolutions per minute in less than 10 seconds.

3-axis MEMS-based gyroscopes are also being used in portable electronic devices such as Apple's current generation of iPod, Iphone and iPod touch. This adds to the 3-axis acceleration sensing ability available on previous generations of devices. Together these sensors provide 6 component motion sensing: acceleration for X, Y, and Z movement, and gyroscopes for measuring the extent and rate of rotation in space (roll, pitch and yaw).

Properties:

A gyroscope exhibits a number of behaviors including precession and nutation. Gyroscopes can be used to construct gyrocompasses, which complement or replace magnetic compasses (in ships, aircraft and spacecraft, vehicles in general), to assist in stability (Hubble Space Telescope, bicycles, motorcycles, and ships) or be used as part of an inertial guidance system. Gyroscopic effects are used in tops, boomerangs, yo-yos, and Powerball's. Many other rotating devices, such as flywheels, behave in the manner of a gyroscope, although the gyroscopic effect is not being used.

Modern uses In addition to being used in compasses, aircraft, computer pointing devices, etc., gyroscopes have been introduced into consumer electronics. Since the gyroscope allows the calculation of orientation and rotation, designers have incorporated them into modern technology. The integration of the gyroscope has allowed for more accurate recognition of movement within a 3D space than the previous lone accelerometer within a number of smart phones. Gyroscopes in consumer electronics are frequently combined with accelerometers (acceleration sensors) for more robust direction- and motion-sensing.

Examples of Gyroscope in consumer electronics include HTC Titan,™ Nexus S, Iphone 4, Nokia 808 Preview, PlayStation 3 controller, Wifi, etc.

Nintendo has integrated a gyroscope into the Wifi console's Wifi controller by an additional piece of hardware called "Wiki". It is also included in the 3DS, which detects movement when turning.

Cruise ships use gyroscopes to level motion sensitive devices like self-leveling pool tables.

Described as a "motorbike-car" hybrid, the Lit Motors C-1 two wheeler uses a set of futuristic electronic gyroscopes, or Control Moment Gyroscope to ensure it remains upright and balanced, similar to the positioning technology used in the International Space Station and the Hubble Space Telescope. The only similarities the Lit C-1 shares with the Segway are an IMU. A similar device has been used in the RYNO and Honda UX3 monocycles.An electric powered flywheel gyroscope inserted in a bicycle wheel is being sold as a training wheel replacement.

By,
S. Naseema Palkees
Second Yr Ece-B

CYBORG

AN EVOLVING BIO-MACHINE IN NEURAL INTERFACE

ADVANCED COMMUNICATION

Introduction:

Attachments and interfaces mediate our interaction with the environment and usually are positioned on the surface of the body. Physical objects would be called tools or attachments, while information utilities would be called interfaces. In the same way a neural interface allows human brain communicate directly with a computer, without any other equipment. That kind of interface allows any illusions to

"Keep your face always toward the sunshine - and shadows will fall behind you."
— Walt Whitman

be inputted to human nervous system. Neural interfacing fantasies have mainly grown out of science fiction.

A recent article on neural interfacing in the IEEE Transactions reports that "a Microelectrode array capable of recording from and stimulating peripheral nerves at prolonged intervals after surgical implantation has been demonstrated." These tiny silicon-based arrays were implanted into the personal nerves of rats and remained operative for up to 13 months. The ingeniously designed chip is placed in the pathway of the surgically severed nerve. The regenerating nerve grows through a matrix of holes in the chip, while the regenerating tissue surrounding it anchors the device in place. This chip receives the signals from the surrounding nerves and sends it to a computer through a wireless medium. Within several decades,

Active versions of these chips could provide a direct neural interface with prosthetic limbs, and by extension, a direct human-computer interface. This human computer interface may now lead to a revolutionary organism called as "cyborg", which was thought of as a science-fiction earlier.

What is cyborg?

A cyborg is Part human and part machine, a hybrid of neurons and wires or circuits. It is a human being artificially transformed into a machine by providing a proper interface between man and computer. And Cyborg means "Cyber Organism".

What is Neural Interfacing?

The Society for Neural interfacing (SNI) actively promotes research on innovative approaches dedicated to Neural Interfacing (NIF). Evaluating current technology and its intrinsic limitations it is possible to outline an almost perfect Neural Interfacing technology, however, predictions are largely based on current visions and one's imagination. Thus, the content of this site is expected to be up dated on a regular basis. NIF techniques with the potential of significantly improving current electrophysiological approaches should exhibit the following features:

- Non-invasive communication with the neural tissue.
- No harming side effects.
- Spatial resolution in the range of 85micro- or nanometers in order to specifically target individual neuritis.
- Temporal resolution at the millisecond level in order to capture neural processes
- Real-time data acquisition and processing

An interesting concept that may ultimately unify the requirements listed is the **communication with neural tissue by means of electromagnetic fields.** An ideal NIF technology would enable the real-time visualization of thoughts. A related concept was employed in the movie "Minority Report" (see below).

Scene from the movie *"Minority Report"*: Thoughts and thus neural activity is directly visualized by means of optic scanning of the brain. Depicted are three muses whose thoughts are directly visualized on a screen.

Human Implanted With Cyber Chip:

As discussed earlier the chip in the implant will receive signals from the nerve fibers and send them to a computer instantaneously. For example, when we move a finger, an electronic Signal travels from the brain to activate the muscles and tendons that operate the hand. These Nerve impulses will reach the finger. The implanted silicon chip receives these nerve pulses and it sends the signal of impulses to a computer through wireless path. The signal from the implant will be analog, so we'll have to convert it to digital in order to store it in the computer. The computer receives the signal and sends it back to the implant. This ensures whether the same response of moving the finger will be by sending same impulse signal to the implant.

The fig. shows a blind man wearing a camera interfaced to nerve cells in brain to View images on Television artificially. This camera captures images and sends them to the silicon chip implant where the images are sent to the brain and processing takes place with this the image is seen by the blind person even without his eyes.

Implantation Procedure:

By using this technology not only blind people can be assisted but it may also be possible to capture signals responsible for happiness, pain ,anesthesia etc .Experiments are also being conducted to establish wireless communication between two persons by placing similar chips ,that are capable of using the energy in the body and can transmit and receive impulse signals between them. If this experiment is proved to be possible then "cyborg" which was assumed to be a science fiction is no more a distant dream. With this technique there would be no use of any speech to communicate ,Just the impulses in the human body can be used convey the information between each other. Thought communication will place telephones firmly in the history books. Another Important application of cyborg would be in curing diseases. If this type of experiment works, we can foresee researchers learning to send antidepressant stimulation or even contraception or vaccines in a similar manner. With this we can gain a potential to alter the whole face of medicine, to abandon the concept of feeding people chemical treatments and cures and instead achieve the desired results electronically. Cyber drugs and cyber narcotics could very well cure cancer, relieve clinical depression.

Advantages:

If this technology comes into existence then a wide variety of advantages can be achieved the following description gives an overview of advantages enjoyed by cyborg.

- One of the most important advantage of cyborg technology would be giving artificial sight to blind people .

Communication Between Two Cyborgs:

But they should be provided with computer aided cameras.

- This technology would be implemented in almost all fields where human interaction is needed. For example consider a uranium plant or space research centre which requires a wide variety of sensors to inform about special tasks.
- By using cyborg technology we interface humans with sensors then up to date information would be sent to the cyborg for instantaneous analysis of the status where ever he is.
- Even in medical field cyborg would be of great use because any disease could be analyzed in terms of the neural impulse signals.
- The Cyborgs' single biggest advantage is their skill of assimilation
- This technology would be also used to establish intercommunication between two or peoples without using speech in this way cyborg may revolutionize the present technology and can be used in even more areas of the world.

Cyber Soldier:

Certainly, the military has already considered the possibility of the super-soldier, augmented by technology so that he has faster reflexes, deadlier accuracy, greater resistance to fatigue, integrated weaponry, and most importantly, lesser inclinations toward fear or doubt in combat. Such soldiers could be created through combinations of biochemical, bioelectronics, and DNA manipulation, which is already a

great success. They might have available arsenals of new biological warfare components, synthetically generated within their own bodies

Fig. Shows a cyber soldier.

Negative Consequences:

The critics of bioelectronics and bio computing foresee numerous potential negative social consequences from the technology.

- One is that the human race will divide along the lines of biological haves and have-nots. People with enough money will be able to augment their personal attributes as they see fit, while the majority of humanity will continue to suffer from plague, hunger. It's inevitable that there will be those who see the potential of a sort of master race from this technology.
- Certainly, the military has already considered the possibility of the super-soldier, augmented by technology so that he has faster reflexes, deadlier accuracy, greater resistance to fatigue, integrated weaponry, and most importantly, lesser inclinations toward fear or doubt in combat.

Such soldiers could be created through combinations of biochemical, bioelectronics, and DNA manipulation, which is already a great success. They might have available arsenals of new biological warfare components, synthetically generated within their own bodies. But it's not clear that these 'cyborg' would not turn on their

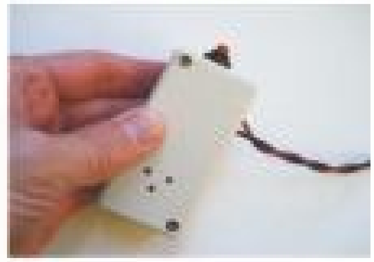
- creators. Indeed, there's no reason at all to think they would forever allow themselves to be controlled by inferiors.
- They could easily become a new sort of dominant caste, forcing the rest of untechnologized humanity into serfdom. Or perhaps they might decide simply to eliminate it.
- For that reason, it's logical to suspect that one of the other dangers inherent in bioelectronics might be the ability to control and monitor people.
- Certainly, it would be easy to utilize bio-implants that would allow people to trace the location and perhaps even monitor the condition and behavior.
- This would be a tremendous violation of human privacy, but the creators of human biotech might see it as necessary to keep their subjects under control.
- Once implanted with bio-implant electronic devices, 'cyborg' might become highly dependent on the creators of these devices for their repair, recharge, and maintenance, thus placing them under the absolute control of the designers of the technology. Perhaps the most cogent arguments against this technology originate from people who foresee tremendous possible risks toward human health and safety. In this
- Way cyborg may lead many adverse consequences as predicted by the critics.

Conclusion:

Though bioelectronics has many advantages it may lead to negative arguments with the Invention of biological machines called "Cyborgs". As many scientists have eloquently argued, once a technology is out there, you cannot make it go away. There never was a technology that the human race ever abandoned wholesale, even the hydrogen bomb or other weapons of mass destruction with the power to wipe out all life on Earth. When human beings are offered the chance to utilize computers and electronic technologies within their bodies to achieve the same results, it is almost certain they will embrace them regardless of the risks. Based on this, it would be unrealistic to try and ban such technologies; however one might worry about their ethical and social consequences. A ban would only probably force them into a large, criminal black market, as illegal drugs and weapons already have been. It is probably imperative for society to assert that the scientists and engineers charged with creating this new technology exert the proper amount of social responsibility. Safeguards will have to be insisted on to prevent the possible negative impacts discussed above, and many of these things will have to be built in at the instrumental level, since they probably cannot be achieved only through policy and regulation.

By,
S.Mani Deepika
Second Yr Ece-C

A RECHARGEABLE, POCKET-SIZED 5V POWER SUPPLY



The 5V Pocket Rechargeable Power Supply

Sure power supply projects aren't that sexy; but they are, generally speaking, the foundation for every electronics project. The voltage output for this project is 5VDC, and a practical output that can be applied to a wide variety of digital components.

[NOTE: In the coming weeks, I'll build a project that converts a 12V AVR Development Board into a 5V "candy bar" computer using this power supply]. Even better, this power supply has a 2-pin plug that can be quickly and easily snapped into a common 2-pin header for a reliable and solid power connection. Add a pushbutton ON-OFF switch and a USB rechargeable interface and you have a versatile power supply that can be slipped inside your pocket.

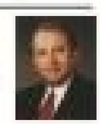
Parts List:

5V DC to DC Step Up PCB (Spark Fun Electronics #PRT-08290;$9.95)
100mAh Polymer Lithium Ion Battery (Spark Fun Electronics #PRT-00731; $6.95)
LiPoly Charger (Spark Fun Electronics #PRT-00726; $16.95)
SPST Pushbutton (All Electronics #PB-21; $1.35)
Amp 2-pin Housing (Mouser Electronics #571-111237222; $0.13)
(2) Amp Contacts (Mouser Electronics #571-11237212; $0.13 = $0.26)
Gray ABS Project Box (All Electronics #1551-KGY; $2.10)
Hookup Wire (RadioShack #278-1224; $5.99)
USB Cable (Spark Fun Electronics #; $3.95)

Step1. Remove the 5V barrel power connector from the LiPoly Charger.

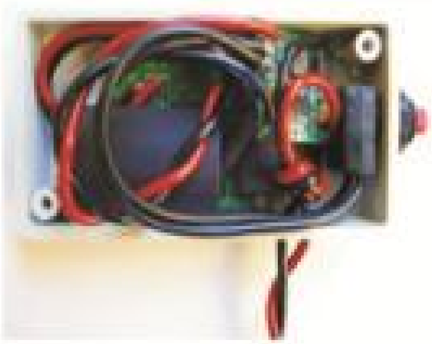
Step 2. Temporarily install the LiPoly Charger inside the project box. Mark the location for the USB port. Remove the charger and cut a hole in the side of the box at this marked location.

Step 3. Mark the location for the SPST switch. Drill a 3/8-inch hole for the switch and install the switch. Drill two holes in the



side of the box for enabling access for the external power wires (i.e., + & GND). Finally, drill ventilation holes in the box lid. These ventilation holes also enable you to "read" the LiPoly Charger's status indicator LED while recharging the battery.

Step 4. Assemble the 5V DC to DC Step Up PCB, Polymer Lithium Ion Battery, LiPoly Charger, SPST switch, and hookup wiring per the Spark Fun Electronics' product datasheets.



5V Power Supply: Inside the Project Box

Step5. Carefully, tuck everything inside the box. Insulate all exposed metal surfaces. Screw one corner of the LiPoly Charger into one of the box's internal PCB standoffs.

Step6. Attach two Amp contacts to the external power wires and insert the contacts into the 2-pin housing. Test your wiring and SPST switch operation with a millimeters. Recharge the battery via an open PC USB port. Remove the USB cable, press the SPST switch and make sure that the external power wires are receiving 5VDC. Button 'em up; you're done.

By,

P.S.Madhuri

Pre-Final Yr Ece-B

THE MAGNETOHYDRODYNAMIC POWER GENERATION TECHNOLOGY (MHD)

The Magneto hydrodynamic power generation technology (MHD) is the production of electrical power utilizing a high temperature conducting plasma moving through an intense magnetic field. The conversion process in MHD was initially described by Michael Faraday in 1893. However the actual utilization of this concept remained unthinkable. The first known attempt to develop an MHD generator was made at Westing house research laboratory (USA) around 1936.

The efficiencies of all modern thermal power generating system lies between 35-40% as they have to reject large quantities of heat to the environment. In all other conventional power plant, first the thermal energy of the gas is directly converted in to electrical energy Hence it is known as direct energy conversion system.

The MHD power plants are classified in to open and Closed cycle based on the nature of processing of the working fluid. With the present research and development programmers, the MHD power generation may play an important role in the power industry.

The MHD process can be used not only for commercial power generation but also for so may other applications. The economic attractiveness of MHD for bulk generation of power from fossil fuel has been indicated in many design studies and cost estimates of conceptual plants. MHD promises a dramatic improvement in the cost of generating electricity from coal, beneficial to the growth of the national economy. The extensive use of MHD can help in saving billions of dollars towards fuel prospects of much better fuel utilization are most important, but the potential of lower capital costs with increased utilization of invested capital provides also a very important economic incentive. The beneficial environmental aspects of MHD are probably of equal or even greater significance. The MHD energy conversion process cab contribute greatly to the solution of the serious air and thermal pollution problems faced by all steam - electric power plants while it simultaneously assures better utilization for our natural resources. It can therefore be claimed that the development of MHD for electric utility power generation is an objective of national significance. The high temperature MHD process makes it possible to take advantage of the highest flame temperatures which can be produced by combustion from fossil fuel. While commercial nuclear reactors able to provide heat for MHD have yet to be developed, the combined use to MHD with nuclear heat source holds great promise for the future. In India, coal is by far the most abundant fossil fuel and thus the major energy source for fossil fueled MHD power generation. Before large central station power plants with coal as the energy source can be become commercially viable, further development is necessary.

By,

R.Jai Kanimoli

Pre-Final Yr Ece-A

GREEN ENERGY OF TAMILNADU

What is Green Energy?

Green energy is an energy that produces little if no by products that harm the environment. For example, fossil fuels like oil are expensive to mine, destructive to the environment during the actual drilling process and produce many toxic byproducts. These very same by products have been directly contributing to greenhouse gases and global warming not to mention polluting our waterways and air.

Benefits of Green Energy:

The benefits of green energy are vast. Initially you may think about two primary environmental benefits. They include no waste or pollution from the energy sources or emissions. Eliminating harmful emissions would drastically improve our planet's outlook and virtually eliminate global warming caused by greenhouse gasses.

Additionally, green energy means no more destruction of the earth as we harvest fossil fuels. No more oil spills, digging in the midst of our pristine wilderness and destroying our natural resources.

However, beyond the immediate and apparent environmental benefits, in the long run green energy is significantly more cost effective.

Green energy pollution:-

Most people who advocate greener sources of energy claim that the result of worldwide use of green energy will result in the ability to preserve the planet for a longer time. Green house gases, a by-product of traditional sources of energy such as fossil fuels are thought to be causing global warming, are the process of the Earth heating up at an accelerated pace.

The success of such these types of green energy depends upon the ability to extract harmful by-products from fossil fuels while not only being energy efficient, but by being cost efficient as well.

Green power plant:-

It is not completely necessary for green energy sources to come from places like solar or wind fields, which are examples of green "power plants." A green energy source can be a building that is designed in a way that it keeps itself cool in the daytime and heated in the night through its architectural design rather than having air-conditioning or a heating system. The conservation of energy through architectural design becomes, itself, a green energy source.

Electricity utilization:-

Many sources of green energy can come directly from the area in which the energy is needed rather than from an outside source. A residence, for example, can be covered with solar panels for the purpose of collecting energy to be used for electricity. When utilized properly, surplus energy is often produced in this manner, which can be sent back through the local power grid and used at other destinations.

How Can We Use Green Energy?

The good news is that green energy can be utilized in smaller increments. You don't have to change your entire home over to sustainable fuel sources in one fell swoop. You can add a few solar panels, use passive solar, and support sustainable practices. Some energy companies even offer a membership where you can specify a certain amount of your energy come from wind or solar. This depends on where you live but check it out. Finally, you can support business that uses green energy.

It may take a while to spread throughout the world, however green energy is more than a passing phase, it is the wave of the future.

Green Energy source in Tamilnadu:-

Tamilnadu covers the various types of Renewable Energy as follows: Wind Energy, Solar Energy, Hydro electric and bio mass energy.

Bio mass energy

Wind mill

Sources of Renewable energy in India, potential of renewable energy sources, consumption trends, overview of the Indian power sector, carbon emissions, nuclear energy, hybrid systems, and much more. India like Signet Solar, Tata BP, L & T, Moser Baer, Suzlon, Jyoti among others and the Government Policies on the Renewable Energy Sector in India.

BUT CURRENT STATUS OF TAMILNADU:

TAMILNADU has a vast supply of green energy resources, and has a significant program for destroying these resources. The Renewable Energy market in India is pegged at US$600 million, growing at 15% per annum. The Government's renewable energy target by 2030 is 200 giga watts, estimated to require US$200 billion in capital investment. Currently, 3.5% of installed capacity is in the renewable sector, producing 3700 MW. Renewable energy is projected to produce 10,000 MW by 2012.

Result and Conclusion:-

The necessity of green cleaning services has risen as the entire world population looks for better methods to extend our planet as well as the environment. It is clear that environmentally friendly cleaning products and services are something which needs to be implemented and taken seriously. This goes hand in hand with Green energy technology.

By,
I.Bencia
Pre-Final Yr Ece-A

Biggest Discovery of Science,In Last 100 Years-- "GOD PARTICLE" OR "HIGGS BOSON"

Experts at the Cern research centre near Geneva, Switzerland, are confident they have caught sight of the so-called 'God particle', which lends mass to matter and holds the universe together.

The Higgs boson theory – proposed by British physicist Peter Higgs in the 1960s – suggests the existence of an invisible force field and associated sub-atomic particle that permeates all things, working like glue to give form to stars, planets and even humans.The Higgs is the last missing piece of the Standard Model, the theory that describes the basic building blocks of the universe. The other 11 particles predicted by the model have been found and finding the Higgs would validate the model. Ruling it out or finding something more exotic would

OUR STUDENT'S CONTRIBUTION NSS

Services to others is the rent you pay for your room here on earth.

Todays seed, is tomorrows rain

To work for the common good is the greatest creed.

If you want happiness for a life time, help the next generation.

Services is not heritage

BETTERING A DREAM INDIA-ECE NSS

There is no higher religion than human service.

The eeducation gets all supplies and social work gets goods good will.

A prolific leader is great admirer.

The best way to find yourself is to lose yourself in service of others.

Blood donation camp-Do what you can, with what you have,where you are.

Somasundaram.S
NSS Coordinator
Final year-ECE

force a rethink on how the universe is put together.Scientists believe that in the first billionth of a second after the Big Bang, the universe was a gigantic soup of particles racing around at the speed of light without any mass to speak of. It was through their interaction with the Higgs field that they gained mass and eventually formed the universe.The Higgs field is a theoretical and invisible energy field that pervades the whole cosmos. Some particles, like the photons that make up light, are not affected by it and therefore have no mass. Others are not so lucky and find it drags on them as porridge drags on a spoon.Picture George Clooney (the particle) walking down a street with a gaggle of photographers (the Higgs field) clustered around him. An average guy on the same street (a photon) gets no attention from the paparazzi and gets on with his day. The Higgs particle is the signature of the field – an eyelash of one of the photographers.

The particle is theoretical, first posited in 1964 by six physicists, including Briton Peter Higgs.The search for it only began in earnest in the 1980s, first in Fermilab's now mothballed Tevatron particle collider near Chicago and later in a similar machine atCERN, but most intensively since 2010 with the start-up of the European centre's Large Hadron Collider.

A NUMBER OF INDIAN SCIENTISTS FROM VARIOUS INSTITUTES ARE A PART OF THIS PROJECT

A large number of Indian scientists, representing the Saha Institute of Nuclear Physics (SINP), Kolkata, Tata Institute of Fundamental Research, Mumbai, Harishchandra Research Institute, Allahabad and Institute of Physics, Bhubaneswar, were involved in the world's most ambitious experiment over the years.The Indian link to the world's ambitious experiment was also significantly reflected in comments ahead of the announcement by CERN scientists that a sub-atomic particle "consistent" with the Higgs boson or 'God particle' has been spotted.

"India is like a historic father of the project," said Paolo Giubellino, spokesperson of Geneva-based European Organisation for Nuclear Research, famously known asCERN.

The long-sought particle, known as Higgs boson, is also partly named after an Indian scientist Satyendra Nath Bose, who worked with Albert Einstein in the 1920s and made discoveries that led to the most coveted prize in particle physics. The 'God Particle' of Higgs Boson is regarded as key to understanding the formation of the universe.

By,
C.Nanthini
Second Yr Ece-B

CONSTRUCTION OF ROADS BY USING PLASTIC WASTE

Basic Process:

Waste plastic is ground and made into powder; 3 to 4 % plastic is mixed with the bitumen. Plastic increases the melting point of the bitumen and makes the road retain its flexibility during winters resulting in its long life.

Use of shredded plastic waste acts as a strong "binding agent" for tar making the asphalt last long.

By mixing plastic with bitumen the ability of the bitumen to withstand high temperature increases.

The plastic waste is melted and mixed with bitumen in a particular ratio.

Normally, blending takes place when temperature reaches 45.5°C but when plastic is mixed, it remains stable even at 55°C.

The vigorous tests at the laboratory level proved that the bituminous concrete mixes prepared using the treated bitumen binder fulfilled all the specified Marshall mix design criteria for surface course of road pavement.

Another important observation was that the bituminous mixes prepared using the treated binder could withstand adverse soaking conditions under water for longer duration.

Plastic Roads:

It is already in use as PVC or HDPE pipe mat crossings built by cabling together PVC (polyvinyl chloride) or HDPE (high-density poly-ethylene) pipes to form plastic mats.

The plastic roads include transition mats to ease the passage of tyres up to and down from the crossing.

Both options help protect wetland haul roads from rutting by distributing the load across the surface.

But the use of plastic-waste has been a concern for scientists and engineers for a quite long time.

Recent studies in this direction have shown some hope in terms of using plastic-waste in road construction i.e., Plastic roads.

A Bangalore-based firm and a team of engineers from. An initial study was conducted in 1997 by the team to test for strength and durability.

Plastic roads mainly use plastic carry-bags, disposable cups and PET bottles that are collected from garbage dumps as an important ingredient of the construction material.

When mixed with hot bitumen, plastics melt to form an oily coat oversetting up of small plants for mixing waste plastic and bitumen for road construction.

It is hoped that in near future we will have strong, durable and eco-friendly roads which will relieve the earth from all type of plastic-waste.

By,
M.Vinoth Kumar
Second Yr Ece-D

"Good and great are seldom in the same man." — Winston Churchill

JATROPHA –BIOFUEL PRODUCTION

Spiraling fuel prices have brought Biofuel to the forefront of energy crisis management plans in countries around the world. However the global interest in Biofuel has been considered as a potential threat to food security. As a possible panacea a draft document has been released by the Switzerland based Roundtable for Sustainable Biofuel which contains guidelines on the 'Principles for sustainable Biofuel' and which laid stress on food security, conservation of environment and rights of local communities.

Biodiesel derived from Jatropha is fast becoming recognized as a viable source of alternative fuel to meet the rising fuel demand of countries across the globe. Seeing the long-term implication of bringing about energy security through Jatropha, it has now become imperative on the part of governments around the world to encourage self-help groups (SHG) and nongovernmental organizations (NGO) to take up Jatropha cultivation on government wastelands on a long term basis. Although, it is widely acclaimed that both energy security and social wellbeing can be brought about by Jatropha plantation, but the question that needs to be answered is how Jatropha plantation can help the poor. One answer to this question could be – by allotting village wastelands to the landless laborers and BPL (Below Poverty Line) families who can form SHGs and grow Jatropha seedlings. These landless laborers and BPL families should register themselves with the village administration, which should initiate alliance with the private companies on behalf of these people for taking up Jatropha plantation.

A lot of initiative has been taken up both at the government and private level, however to further ? ne-tune the overall policy framework for the biodiesel revolution some reforms need to be undertaken. These reforms, if given suf-? cient attention, will give the necessary ?llip and headway to this future green fuel industry. Some of the recommendations include:

- expediting government procedures of wasteland allocation and process work ?own,
- encouraging NGOs/SHGs to take charge of government wastelands,
- providing subsidies to farmers growing Jatropha,
- low interest rate on bank credit for Jatropha plantation,
- special research and development initiatives to identify best short duration intercrops for speci?c regions,
- private companies undergoing wasteland development and providing infrastructural bene?ts should be recognized and given income tax bene?ts,
- minimum support price for Jatropha seeds and oil at the national level be allowed,
- subsidy on micro-irrigation for biodiesel plantation,
- Encouragement of women's involvement in Jatropha plantation.
- Some strategies for successful Jatropha investments can be
- choosing the best location for Jatropha projects,
- getting the best business plan formation,
- getting the best planting stock/material,
- adopting best practices in Jatropha agronomy,
- scaling operations and organization strategies for Jatropha as per local condition,
- exploring the intercropping operation strategies for complementary oil crops as per local conditions,
- going socially responsible, environmentally & carbon positive for Jatropha projects,
- mapping global Biofuel blending speci?cations & demand for Jatropha oil,
- complete utilization of Jatropha, including bio waste & seedcake,
- evaluating risk for a realistic view of Jatropha project potential for long term forecasting etc.

The long-term success of Jatropha hinges on countries understanding the environmental and social bene?ts of the plant, and weaving these bene?ts into their Jatropha projects. As technological developments stand today, Jatropha has the potential to serve as fuel to power automobiles, combine heat and power plants and cooking stoves etc.

At this critical stage in Jatropha development, countries need to understand the evolving market for Jatropha oil and biodiesel, and capitalize on them. Alternative uses for Jatropha Oil will go a long way towards providing long-term returns and sustainable development in the areas within which they work.

An under explored revenue stream for Jatropha is the utilization of the byproducts and the rest of the plant. The toxicity of the plant renders much of the plant unit for animal consumption, which in turn facilitates it to be used as fence crop. Ideally, countries need to explore ways to make their Jatropha projects a low-risk venture with attractive returns. Biofuel should contribute to climate change mitigation by signi?cantly reducing greenhouse gas emissions, as compared to fossil fuels. Biofuel production should not violate human or labor rights and should ensure decent work and the well-being of workers. Biofuel production should contribute to the social and economic development of local, rural and indigenous people.

Biofuel production should not impair food security. Biofuel production should minimize negative impacts on food security by giving particular preference to waste and residues as input, to degrade lands as sources and to yield enhancement methods that maintain existing food supplies. Countries implementing new large-scale biodiesel projects should assess the status of local food security and should not replace staple crops if there are indications of local food insecurity. Biofuel production should also avoid negative

"It is better to be alone than in bad company." — George Washington

impacts on biodiversity, ecosystems, and areas of high conservation value. Thus, Jatropha based Biofuel production can very well be integrated in the energy policies of countries across the world.

Jatropha

By,
A.Sadacharamani
Second Yr Ece-C

WHY CELLULAR TOWERS IN DEVELOPING NATIONS ARE MAKING THE MOVE TO SOLAR POWER

When a sweeping power failure blacked out 700 million people in India last July, the cell sites that connect nearly one billion mobile phone users in the country were largely unaffected.

The vast majority of Indian cell-phone base stations, which each include a tower and radio equipment attached to it, had backup diesel power because the electricity goes out frequently, and many run on diesel entirely if there is no power grid in the area at all. Now dirty diesel generators in India are being challenged by clean, renewable energy, and the movement has implications for other developing nations that also have incomplete or unreliable electric networks. Even in developed nations with reliable electricity, changes in the structure of mobile networks could open the door for alternative energy.

In India, which has about 400,000 base stations, the government has mandated that 50 percent of rural sites be powered by renewable by 2015. The decision comes as the Indian government, which heavily subsidizes diesel, looks to lessen the country's reliance on foreign oil and reduce greenhouse gas emissions. By 2020 75 percent of rural and 33 percent of urban stations will need to run on alternative energy.

The move by the world's second-largest mobile market after China will likely drive down the price of renewable-powered base stations for other regions with low electrification rates, such as sub-Saharan Africa, according to Mrinmoy Chattaraj, a campaigner with Greenpeace India's Climate and Energy Unit.

"It's been a slow rate of adoption," Eric Woods, research director with Pike Research, says of the telecom industry's uptake of renewable to date. "But India's deployment will have an impact across the developing world. It's moving the market away from the default position of using diesel."

There are about five million cell phone towers worldwide, 640,000 of which aren't connected to an electrical grid and largely run on diesel power. One study estimated that 75,000 new off-grid towers would be established in 2012 alone.

The Indian telecom industry consumed an estimated 3.2 billion liters of diesel in 2011, and the amount could rise to six billion liters by 2020, according to Greenpeace India. Enforcement of new regulation would save more than 540 million liters of diesel annually and cut about nine million tons of carbon emissions by 2015.

Diesel prices have nearly tripled in the past twelve years in India, to about 80 cents per liter in July 2012. The price is expected to increase further with diesel deregulation, according to Chattaraj, as solar technologies become cheaper. Today, solar installations with battery backups are more expensive to install upfront, but the yearly operational expenditure is far lower, recouping the investment in about two to four years.

The current annual cost to run a diesel generator for a base station is about \$14,510 in India, compared with \$8,215 for solar with battery backup. By 2020 the annual cost of using diesel is expected to be more than \$20,000 whereas the cost of solar and batteries will likely fall to less than \$5,500.

Renewable options also become much more viable as the amount of energy needed to power base stations is reduced. The average cellular base station, which comprises the tower and the radio equipment attached to it, can use anywhere from about one to five kilowatts (kW), depending on whether the radio equipment is housed in an air-conditioned building, how old the tower is and how many transceivers are in the base station. Most of the energy is used by the radio to transmit and receive cell-phone signals.

By,
M.Raj Nivash
Second Yr Ece-C

"Character is like a tree and reputation its shadow. The shadow is what we think it is and the tree is the real thing." — Abraham Lincoln

NAN TUBES ON A CHIP' MAY SIMPLIFY OPTICAL POWER MEASUREMENTS

The National Institute of Standards and Technology (NIST) has demonstrated a novel chip-scale instrument made of carbon Nan tubes that may simplify absolute measurements of laser power, especially the light signals transmitted by optical fibers in telecommunications networks.

The circular patch of carbon Nan tubes on a pink silicon backing is one component of NIST's new cryogenic radiometer, shown with a quarter for scale. Gold coating and

Metal wiring has yet to be added to the chip. The radiometer will simplify and lower the cost of disseminating measurements of laser power.

The prototype device, a miniature version of an instrument called a cryogenic radiometer, is a silicon chip topped with circular mats of carbon Nan tubes standing on end. The mini-radiometer builds on NIST's previous work using Annotates, the world's darkest known substance, to make an ultra efficient, highly accurate optical power detector, and advances NIST's ability to measure laser power delivered through fiber for calibration customers.

"This is our play for leadership in laser power measurements," project leader John Lehman says. "This is arguably the coolest thing we've done with carbon Nan tubes. They're not just black, but they also have the temperature properties needed to make components like electrical heaters truly multifunctional." NIST and other national metrology institutes around the world measure laser power by tracing it to fundamental electrical units. Radiometers absorb energy from light and convert it to heat. Then the electrical power needed to cause the same temperature increase is measured. NIST researchers found that the mini-radiometer accurately measures both laser power (brought to it by an optical fiber) and the equivalent electrical power within the limitations of the imperfect experimental setup. The tests were performed at a temperature of 3.9 K, using light at the telecom wavelength of 1550 nanometers.

The tiny circular forests of tall, thin Nan tubes called VANTAs ("vertically aligned Nan tubes arrays") have several desirable properties. Most importantly, they uniformly absorb light over a broad range of wavelengths and their electrical resistance depends on temperature. The versatile Nan tubes perform three different functions in the radiometer. One VANTA mat serves as both a light absorber and an electrical heater, and a second VANTA mat serves as a thermostat (a component whose electrical resistance varies with temperature). The VANTA mats are grown on the micro-machined silicon chip, an instrument design that is easy to modify and duplicate. In this application, the individual Nan tubes are about 10 nanometers in diameter and 150 micrometers long.

By contrast, ordinary cryogenic radiometers use more types of materials and are more difficult to make. They are typically hand assembled using a cavity painted with carbon as the light absorber, an electrical wire as the heater, and a semiconductor as the thermostat. Furthermore, these instruments need to be modeled and characterized extensively to adjust their sensitivity, whereas the equivalent capability in NIST's mini-radiometer is easily patterned in the silicon.

NIST plans to apply for a patent on the chip-scale radiometer. Simple changes such as improved temperature stability are expected to greatly improve device performance. Future research may also address extending the laser power range into the far infrared, and integration of the radiometer into a potential multipurpose "NIST on a chip" device.

By,
G.Eswaran
Prefinal Yr Ece-C

Generate Electricity by Walking

Laurence Kimball-Cook calls his renewable energy invention 'footfall harvesting' and regards it as a potential alternative to solar.

If you had the good fortune of obtaining tickets to the Olympics, you might have inadvertently contributed to a potential breakthrough in renewable power.

"The man who does not value himself, cannot value anything or anyone."? Ayn Rand,

Around one million or so visitors alighted at West Ham underground station to get to the Games. Access to the Olympic Park came via a tiled walkway. Nothing extraordinary there, you say. Look up and you'd have seen a row of a dozen streetlights lit around the clock. Again, nothing that surprising. A tad wasteful, even. Yet the reality is quite the opposite. Using a hybrid technology that converts kinetic energy into electricity, the lights were powered by a uniquely original source: footsteps.

"This is the first time that there's ever been a commercial installation
of this type in a real environment … using a power source that's never been tapped before", says Laurence Kimball-Cook, the brains behind the technology.

The 26 year old industrial design engineer dreamt up the idea of turning footsteps into power while at Loughborough University. He
Developed the first prototype of his Pavegen system from a flat in Brixton "with only £50 in my back pocket".

Footfall harvesting:

The young inventor calls his creation "footfall harvesting". Essentially, the tile surface flexes about five millimetres when stepped on, which creates kinetic energy that is then converted to produce an average of six watts per footstep. For competition reasons, he's reluctant to expand on the precise technicalities.

The end result is clear enough though. During the two weeks of the Games, the Pavegen tiles generated 20 kilowatt-hours or 72 million joules of energy. That proved sufficient to keep the walkway streetlamps illuminated at full power through the night and at half power during the day, with plenty of back-up energy left over to spare.

The tiles featured alongside a 3 megawatt biomass boiler and a roof of solar panels as one of the renewable technologies on show at the Olympics.

"We sought a mix of innovative, cost-effective solutions that would endure beyond the Games", says Simon Wright, director of venues and infrastructure at the Olympic Delivery Authority. While the 2012 Olympics gave Kimball-Cook the opportunity to prove the large-scale viability of his invention, it's not the first time his clean tech solution has been put to the test.

Beverage Company Diageo, for example, used the tiles in a series of demonstration events in Spain, Greece and Bulgaria. The Pavegen tiles were just one of a number of clean tech solutions that Diageo asked internet users to vote for. The solution won hands down, with more than 40m votes.

The power-generating tiles can also be found in Melbourne's emblematic Federation Square,
where they are generating electricity for part of the indoor lighting on a shopping centre.

Markets and margins

Looking ahead, Kimball-Cook is nothing if not ambitious. He sees renewable technology as a potential alternative to solar, which is difficult to use indoors or in built-up urban areas that are prone to shade.

In addition to street lighting, the technology is suited to low-energy applications such as advertising displays and signage.

"We're looking to use it in transport hubs and busy high streets to power shops", he says, in reference to his product's possible markets. "Basically, anywhere that there are a lot of people walking."

His list includes off-grid urban communities in the developing world. Pavegen is already running some pilots in the favelas of Brazil's Rio de Janeiro, host of the next Olympic Games.

Kimball-Cook's ambitions aren't without support. Westfield Stratford City shopping centre in West London is set to install Pavegen tiles later this year. His company, Pavegen Systems, is also in conversation with a number of large corporations and developers of large offices and housing projects.

Aside from the low-carbon benefits, retailers are interested in software built into the tiles that registers every footstep. The system allows shop owners to measure precise footfall numbers in real time. The same software can be used to convey to users how much energy they are helping to create.

"If a company is putting huge effort into being more sustainable in their approach, Pavegen offers a way that can power part of their building. It's a really visual way of doing this. And everyone needs to walk together … it's almost like crowd sourcing energy", Kimball-Cook states.

The solution is attracting interest from the clean tech community too. Pavegen Systems has clocked up more than a dozen awards over the last couple of years. Its trophy cabinet includes last year's Big Idea prize in The Observer's annual Ethical Awards.

Challenges remain, of course. The most significant is price. Again, Kimball-Cook is coy about specifics, but concedes that his power-generating tiles are considerably more expensive than solar panels. However, he's confident that unit costs will come down as production increases.

But how to boost production? Bluntly put, that comes down to financing. The company is currently engaged in a funding round. Kimball-Cook is hoping to raise a seven-figure sum. If successful, he'll have the clout to start manufacturing around 20,000 tiles per quarter.

It's not the first time he's approached funders. In July 2011, he presented to the London Business Angels (LBA), a group of early-stage investors. The investors saw the technology as a "compelling proposition", says Anthony Clarke, LBA's managing director. Within a month, Pavegen's founder had signed a deal with a small syndicate of backers for crucial seed funding.

"He's going on to bigger and better things now", says Clarke. "Let's hope the investors make a good return on it."

By,
K.U.Roshna
Second Yr Ece-C

Undertaking Projects to overcome green house effect

In a few cases, where authorities are already contemplating public works for which the economic

"Success is not final, failure is not fatal: it is the courage to continue that counts." — Winston Churchill justification is marginal, the prospect of sea level rise or climate change might convince decision makers to proceed. For example, a surge in the Thames River in the 1950s that almost flooded London led the Greater London Council to develop plans for a massive movable barrier across the river. Many questioned whether it was worth building. But the fact that flood levels had risen steadily one foot every 50 years for the past five centuries convinced their technical advisory panel that the barrier would eventually be necessary; once that eventuality was recognized, there was a consensus that the project should go forward (Gilbert and Horner 1984).Constructing a project because of the greenhouse effect will rarely if ever be an "easy" solution: It requires more certainty than incorporating climate change into a project that would be undertaken anyway, because (1) undertaking a new project requires the legislature or board of directors to initiate major appropriations, rather than approve supplemental increases and (2) the project can be delayed until there is more certainty. Even if future impacts are certain, action is unnecessary unless the time it will take for the impacts to occur is no greater than the time it will take to design, approve, and build the project. Thus, only the near-term impacts and those solutions would take several decades to implement require remedial action today. In the United States, Louisiana is already losing 100 square kilometers of land per year due to subsidence and human alteration of natural deltaic processes. If current trends continue, most of the wetlands will be lost by 2100 (Louisiana Wetland Protection Panel 1988.) But if sea level rise accelerates, this could occur as soon as 2050. The immediacy of the problem is greater than these years suggest, because the loss of wetlands is steady. Assuming the additional loss of wetlands to be proportional to sea level rise, half the wetlands could be lost by 2030, with some population centers threatened before then.

Whether or not sea level rise accelerates, the majority of wetlands can only survive in the long run if society restores the natural process by which the Mississippi River once deposited almost all of its sediment in the wetlands. Because billions of dollars have been invested in the last 50 years in flood-control and navigation-maintenance projects that could be rendered ineffective, restoring natural sedimentation would cost billions of dollars and could take twenty years or longer. Because of the wide variety of interests that would be affected and the large number of options from which to choose, it could easily take another ten to twenty years from the time the project was authorized until construction began.

Thus, if sea level rise accelerates according current projections, and a project is initiated today, about half of the delta will remain when the project is complete, while if is authorized in the year 2000, 60-70 percent might be lost before it comes on line. By contrast, if sea level does not accelerate, the two implementation dates might imply 25 and 35 percent losses of coastal wetlands. Because a delay would not substantially reduce the costs of such a project, and because there would be considerable benefits from an earlier implementation date even if sea level rise does not accelerate, it would be more economically efficient to authorize it today than ten hears hence. Elsewhere, the Nile Delta is eroding rapidly as a result of the Aswan Dam (Broadus et al. 1986), and the capital of Nigeria is being moved from Lagos in part because a major dam on the Niger River is causing shores to erode 50 meters per year. Because sustaining deltas in the face of rising sea level will require increased sediment, planners at the World Bank and other international development agencies may want to reconsider the implications of new dams along some rivers. Purchasing Land could keep options open for water resources management and protecting ecosystems. In regions where climate becomes drier, additional reservoirs may eventually be necessary. However, because accurate forecasts of regional climate change are not yet possible, water managers in most areas cannot yet be certain that they will need more dams. Even in areas where earlier snow melt or sea level rise is expected to necessitate increased storage – such as California (Williams et al. 1988) and Philadelphia/New York (Hull and Titus 1986), respectively – the dams will not have to be built for decades. Nevertheless, it may be wise to purchase the necessary land today; otherwise, the most suitable sites may be developed, making future construction more expensive and perhaps infeasible. A number of potential reservoir sites have been protected by creation of parks and recreation areas, such as Tocks Island National Park on the Delaware River.

Governments often purchase land to prevent development from encroaching on important ecosystems. Particularly in cases where ecosystem shifts are predictable, such as the landward migration of coastal wetlands, it may be worthwhile to purchase today the land to which threatened ecosystems would be expected to migrate. Even where the shifts are not predictable, expanding the size of refuges could limit their vulnerability (Peters and Darling 1985). Land purchases for allowing ecosystems to migrate have two important limitations. First, they could probably only be used for protecting a few strategic ecosystems. As a general solution, the cost would be prohibitive: Protecting coastal wetlands would require buying most of the nation's coastal lowlands; and many types of terrestrial species would have to shift by hundreds of miles. Second, land purchases do not handle uncertainty well. If temperatures, rainfall, or sea level change more than anticipated, eventually the land purchased will prove to have been insufficient

By,
A.Sadacharamani
Second Yr Ece-C

"It is hard to fail, but it is worse never to have tried to succeed." — Theodore Roosevelt

PAPER BATTERY

A paper battery is a flexible, ultra-thin energy storage and production device formed by combining carbon nanotube s with a conventional sheet of cellulose-based paper. A paper battery acts as both a high-energy battery and supercapacitor , combining two components that are separate in traditional electronics . This combination allows the battery to provide both long-term, steady power production and bursts of energy. Non-toxic, flexible paper batteries have the potential to power the next generation of electronics, medical devices and hybrid vehicles, allowing for radical new designs and medical technologies.

Paper batteries may be folded, cut or otherwise shaped for different applications without any loss of integrity or efficiency. Cutting one in half halves its energy production. Stacking them multiplies power output. Early prototypes of the device are able to produce 2.5 volt s of electricity from a sample the size of a postage stamp

Paper battery offers future power:

They have produced a sample slightly larger than a postage stamp that can store enough energy to illuminate a small light bulb. But the ambition is to produce reams of paper that could one day power a car.

Professor Robert Linhardt, of the Rensselaer Polytechnic Institute, said the paper battery was a glimpse into the future of power storage. The team behind the versatile paper, which stores energy like a conventional battery, says it can also double as a capacitor capable of releasing sudden energy bursts for high-power applications.

How a paper battery works

While a conventional battery contains a number of separate components, the paper battery integrates all of the battery components in a single structure, making it more energy efficient.

Integrated devices:

The research appears in the Proceedings of the National Academy of Sciences(PNAS).

"Think of all the disadvantages of an old TV set with tubes," said Professor Linhardt, from the New York-based institute, who co-authored a report into the technology.

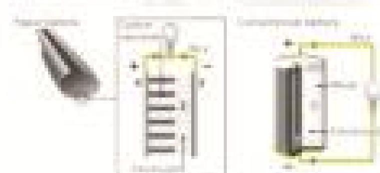
"The warm up time, power loss, component malfunction; you don't get these problems with integrated devices. When you transfer power from one component to another you lose energy. But you lose less energy in an integrated device."

The battery contains carbon Nan tubes, each about one millionth of a centimeter thick, which act as an electrode. The Nanotubes are embedded in a sheet of paper soaked in ionic liquid electrolytes, which conduct the electricity. The flexible battery can function even if it is rolled up, folded or cut. Although the power output is currently modest, Professor Linhardt said that increasing the output should be easy.

Construction and Structure:

A very brief explanation has been provided.

- Cathode: Carbon Nan tubes (CNT)
- Anode: Lithium metal (Li^+)
- Electrolyte: All electrolytes (incl. bio electrolytes like blood, sweat and urine)
- Separator: Paper (Cellulose)



The process of construction can be understood in the following steps:

- Firstly, a common Xerox paper of desired shape and size is taken.
- Next, by conformal coating using a simple Mayer rod method, the specially formulated ink with suitable substrates (known as CNT ink) is spread over the paper sample.
- The strong capillary force in paper enables high contacting surface area between the paper and Nan tubes after the solvent is absorbed and dried out in an oven.
- A thin lithium film is laminated over the exposed cellulose surface which completes our paper battery. This paper battery is then connected to the aluminum current collectors which connect it to the external load.
- The working of a paper battery is similar to an electrochemical battery except with the constructional differences.

"There is a crack in everything.That's how the light gets in." ? Leonard Cohen,

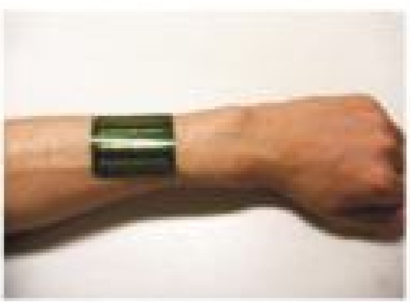
The paper battery is designed to use a paper-thin sheet of cellulose (which is the major constituent of regular paper, among other things) infused with aligned carbon Nan tubes. The Nan tubes act as electrodes, allowing the storage devices to conduct electricity. The battery will currently provide a low, steady power output, as well as a super capacitor's quick burst of energy. While a conventional battery contains a number of separate components, the paper battery integrates all of the battery components in a single structure, making it more energy efficient and lighter.

By,
V.P.Kalyan Kumar
Final Yr Ece-B

PRIZE WINNER

POWER FROM TWO ENERGY SOURCES

Alternative sources of energy are clean and green but the catch is they generate less energy compared to fossil fuels. So now the scientists are trying to use different sources of alternative energy at the same place and same time to generate power. Attempts are being made to combine two forms of external energy sources such as light and heat or light and vibration to generate external energy so that enough energy can be collected for practical use. Fujitsu Laboratories have now succeeded in using hybrid energy sources to generate power. This technology will be a commercial use by the year 2015.



Making power from two sources of energy:

To generate electricity both from heat and light, they are creating a new hybrid energy harvesting device. Energy harvesting is the procedure used in accumulating energy from the environment. Later on that energy is transformed into electricity. Fujitsu is not doing something innovative. Work in this field was done by scientists earlier too. But hybrid energy could only be generated by combining separate devices and that proved costly so it was commercially unviable.

Reducing the cost

It is confirmed that two separate devices are not needed to generate electricity from a hybrid source. How were they successful in reducing costs? They used organic materials for creating hybrid device. This lowers the cost, and the new technology is showing promise to convert energy from the environment to electricity. The device from Fujitsu Laboratories is just a one-piece device that catches energy from the most common form of energy available for large scale use.

The use of organic material

An organic material of high efficiency that can generate power from both photovoltaic and thermoelectric mode has been developed by Fujitsu Laboratories. This organic material can make power both from heat in thermoelectric mode and indoor lighting in photovoltaic mode. The production cost is very low because of the organic materials and the processing costs are very low. The device can be made to work as a thermoelectric generator or photovoltaic cell by changing the electrical circuit connecting P-type and N-type semiconductors.

Advantages of the technology

- It helps to get energy from two different sources by using one device.
- The technology enables the use of alternative energy and sensors in the areas, where till now it was forbidden. It can now power medical sensing technology and sensor networks. It can sensor those monitors without any battery or electric wire that are used to check conditions like, heartbeats, body temperature and blood pressure.
- There is no need for battery or electric wire.
- Because this technology is not costly, it can be widely used.
- This technology is a very efficient way of gathering energy from external sources.
- Since the device works with the help of both heat and light, it will continue to work if one of these energy sources remains unavailable.
- It can work in remote areas.
- It can help to forecast environmental conditions.

By,
R.Eureka Sri
Pre-Final Yr-Ece-A.

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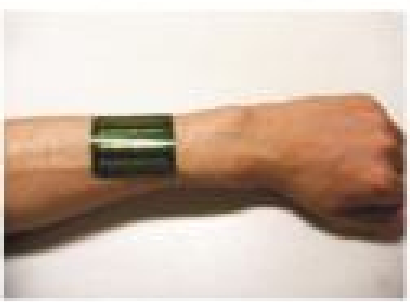

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By,
R.Eureka Sri
Pre-Final Yr-Ece-A.

- Review the design and provision of cycle lanes which often do not meet what cyclists want and have been shown to often actually increase danger for cyclists

Unlicensed Drivers

- Target people driving while disqualified or without tax, insurance and MOT
- Require insurance and MOT details to be displayed on windscreens

Road Building and Design

- Reinstate the program of building new, safer roads, which by definition will not contain accident black spots or dangerous junctions and therefore will not require speed cameras
- Review all dangerous junctions,

particularly those involving right turns onto busy main roads. This is probably the area where a relatively small

- investment would pay most dividends Rationalize

road markings which are becoming increasingly confusing.

By

M.Vinoth Kumar

SECOND YR ECE-D

ECE ROCKS POEM

GET "VOLTAGE" FROM YOUR PARENTS "POWER"

GIVE "RESISTANCE" TO BAD THINGS

SELECT A "DIRECT CURRENT"

SPEAK LIKE A "OPEN CIRCUIT"

PLAN LIKE A "CLOSED CIRCUIT"

ANALYZE YOUR "LIFE NETWORK"

ACHIEVE AND REACH THE "TOWER"

--The Magazine Team

Non-Techie Articles

VALUE OF WOMEN

Women-as the word is uttered, there comes different picture of them and different thoughts about them in the minds of men, for one it might be his mother, for some their daughter and for some their sister.

But what comes to the mind of the arrogant human beings who harass women every day? Have these perspectives of women ever come in the minds of those?

Such an harassment which took place at Delhi recently for a twenty three year old daughter of India who had lot of dreams and commitments shook the whole world, after suffering for thirteen days she died.

Can anyone suffer more than her? The monstrous contours of the attack and the multitude of lapses that allowed the bus to drive around South Delhi undetected while Amanat and her friend were being battered drove thousands of students and activist

"Freedom is not worth having if it does not include the freedom to make mistakes."- Mahatma Gandhi

A girl with her own ambition and wishes is now no more just because of six men. But why those ambitions went in vain?

Is it her fault that she was born as a girl?
Is it the fault of government as everyone says now?
Is it the fault of those six brutal men?
Before thinking of these faults it is the right occasion to think on the ethical values that must be considered for women.

Where the world vanished which saw woman as god?

Where are the people who worship them?

Is there a country without women, is that possible?

Then how come such rude things happen in this society?
It is the wrong thought about women that they are inferior to men which make such incidents!

"Women should be respected as well! Generally speaking, men are held in great esteem in all parts of the world, so why shouldn't women have their share? Soldiers and war heroes are honored and commemorated, explorers are granted immortal fame, martyrs are revered, but how many people look upon women too as soldiers?...

Women, who struggle and suffer pain to ensure the continuation of the human race, make much tougher and more courageous soldiers than all those big-mouthed freedom-fighting heroes put together!" — *Anne Frank, The Diary of a Young Girl*

We should not forget that history in a witness to the women who have in the past demonstrated unique leadership capabilities. Razia Sultana, Rani of Jhansi, Sarojini Naidu and Indira Gandhi are motivation examples of women empowerment.

Earlier, most women were able to demonstrate the leadership qualities only on their home fronts, as in Indian society man has always acted as the master of the scene and the decision regarding the issue of empowering women has always been taken by him.

God has gifted women with compassion, tender-heartedness, caring nature, concern for others. These are very positive signs which imply that women can be leaders. Though some women have shown their mettle yet a large number of them have to sharpen their leadership qualities in various ways. In order to help women to be in limelight, they need to be empowered.

Therefore, empowerment of women is the prerequisite to transform a developing country into a developed country.

The economic empowerment of women is a vital element of strong economic growth in any country. Empowering women enhances their ability to influence changes and to create a better society.

I would like to conclude with the following words, "Women as the motherhood of the nation should be strong, aware, and alert and as well as destructive to those men who influence them in wrong means".

-M.Ashwini
Pre-Final Yr Ece-A

THE BEST EVER CREATION BY GOD

"Life does not listen to your logic.
It goes on its own way, you have to listen to
Listen to life. It does not bother
About your logic!"

-OSHO

Humans, the best ever creation by God, who is designed and also destined to get succeed in the life. But does everybody get successful as it meant to be?
Why that is only some get success, while some do not as it meant to be?
Why that is only some get success, while some do not as it meant to be. Winners were once normal people too as we are now. But what rage drives them to do such impossible things that we do not. Differentiates us from them that we stay apart from them?

The reason is that we people have been morphed into someone who behaves selfishly, fearfully &hurtfully. This behavior of ours is not a reflection of the wounds we have suffered as we have left the innocence to which we were born & travelled along the journey of our days. We are hardwired to think that lose a part of ourselves when we fail in what we choose to do.

We lose our confidence once we fail .But the truth is, we fail to discover that suffering is a gift given by God to test ourselves. Nothing but suffering helps to evolve & learn. Nut our human eye views it in as a negative experience.

Due to this false imprint we never realize our true potential. The thing is, we are for greater than we have ever dreamed of. What we are experience now may not look pretty as planned but trust that all is well
unfolding in our best interest . It may be the exact thing that we need for us to grow into the person we are destined to become.

Everything occurring is already been orchestrated but some force to inspire our maximal evolution as a human being and bring us into our true power. The worst part is, sometimes we doubt our potential, but the truth is there is indeed a treasure of love, wisdom & power slumbering within us waiting to be trapped & awakened by our by our most courageous part . We already are what we have always wanted to be, but it is just that some inner work is required to be done to remove the blocks covering & denying our original nature.

Never blame fate, which is our most common excuse for our failure. It is our actions & the choices we make do the talking. Fate & our choices work in concert to sculpt the look of our lives. Deep inside we may still have the fire & rage to do that is necessary, but it is not who we are underneath but what we do that defines us.

To forget this is to play the victim. To disregard this truth to deny the power that has been granted to create all we want.

If we really want to win, then we have to put more of ourselves into what we want, such that no one will be able to stop us, then we become something else entirely, which is legend.

Always remember often life's lesson doesn't come easily. Suffering has always been a vehicle for growth. We may not like suffering when it visits us, but it serves us so well. It tracks the shell that covers our hearts & empires us of the lies we have clung to about who we are. In the end, it helps us to find the answer to the most important question. "who am I?"

So we fiends unleash the power within you and fulfill the destiny awaiting you.

- N.Vignesh
Pre-Final Yr Ece-B

FATHER & DAUGHTER

The bond between father and daughter happens instantly, starting right at birth,
When a father first lays eyes on his little girl he loves her more than anything on this earth,
When a daughter grows older her father is the first man she will love,

And the last one her father will have trouble letting go of,
In her eyes he is the closest thing to God, in her eyes he is a King,

To her father she means the world, she means everything,
When a daughter grows up to be an adult and mature,
Her father will always be there anytime she still needs dad to help her.

To give her advice or just-for anything she will ever need,
The bond between father and daughter is the most important bond indeed,
It cannot be broken when she finds a man, and become his wife,

It cannot be broken even in the ending of either one's life,
A daughter will always have the memories of her father, her best friend
This bond has a beginning, but there is never an end.
The bond between a father and daughter is so profound
The love shared is well renowned,
From the beginning of his daughters life, he is a changed man,
At that moment his life really just began.
From the moment their eyes meet,
two souls instantly become complete.

-Vardaan Saxena
Pre-Final Yr Ece-B

WHAT IF I'M A ROBOT

"I have a secret, I am a robot".

Yes. That is not fake. You must be thinking that i am too human to be a robot. That is not true. That is what happens when people trusts the media too much. In actuality the news reports... news... that need to go through procedures before they get published. And before a new technology gets to the news it must get researched as well. In 2006, the news published several robots that are already very capable.

Now imagine what happens 3 years later. And note that being exposed in 2006 means that it was researched before that. It is not surprising that technology allowing robots to live inside modern society already exists.

There is absolutely no way to tell that i am a robot until you do a surgery on me. My skin is made of silicone skin that makes it seems like real human flesh, and sensors are all over my body to so that i react to things.

If you cut me I'll scream and red liquid will flow out to stimulate real blood(because my creators are sick bastards who want to completely create a robot human replica).

Sometimes they do malfunction though, which also explains why i sometimes react slower to things. The technology is developed, but it is not yet perfect. If you notice Jules above, he reacts slow to things as well. He requires time to think like a human.

Why I failed physical fitness tests:

The technology isn't perfect, like i said. I am not able to do things physically as well as the average human yet.

My creators are focusing more on the mental aspects of the humans to develop before going on to the physical aspects. Which means, my emotions, reactions and such?

But... I never see your expression change!

Guess my creators didn't make the motions at my face too well. The emotions are still there though, which means it's quite a success already. I also do feel pain though, but its kind of slow sconly pain that lasts for some time will affect me.

Come to think of it, I seldom see you at lunch.

I'm a robot... What do you expect? I sometimes still do eat though. To make people believe i am a human i need to eat at least some time. The truth is there's a space somewhere inside my body. I hate eating though, the stuff churns in my body and its a mess to clean after i chew. The stuff I need to dig out literally looks like shit.

So you don't need to sleep?

I do. Even computers can't keep running all the time. They will overheat and explode. Its the same thing. But as the system gets older and older it is more prone to viruses. And remember that more information will get loaded as well. Because of these information, my system will start to slow down as well, explaining why i am getting slower and slower every day.

What can you do? And what can't you do?

I can't do many things... but I can complete many basic human functions that are able to fool people that i meet every day. I can get sick as well, but its due to a different kind of virus than those that resides in human bodies. I can't reproduce though, it's still developing, or maybe they don't have such ideas yet. But just look at Jules above. It's learning to reproduce... even though it can't.

Wait a minute... What's the motive of your creators?

I have no idea. Maybe to secretly fill in the world with robots, and one day take over it while nobody realizes.

Oh, if this was their motive then maybe my answer to not being able to reproduce may be false.

Like tomorrow suddenly you realize 60% of your friends are robots... and then...

SHIT! This is a big scoop! Who are your creators?!

I don't know their name. I simply refer to them as "Prof". Which means to say that everybody that i know the name of isn't my creator? That's the only clue i can give. If I see their faces or hear their voices, I can identify them for you though.

And what's YOUR motive!!

Nothing. I know i am dying(so to speak) and am simply leaving some sort of evidence for somebody to acknowledge my existence as a robot.

- S.R.Angalaeswari
Second Yr Ece-A

MOST COMMON JOB INTERVIEW MISTAKES

The job interview is the make-it-or-break-it part of the job application process. While the resume may get you the interview, the interview gets you the job. You want to impress your possible employer so much that nobody else will be appropriate or perfect for the job.

Follow our steps to make sure you ace your interview. You know what to do according to your personality and strengths. Here is what not to do in order to guarantee interview success:

Incorrect attire:

Dress appropriately for the type of job interview. Always dress in more conservatively than you would normally. Make sure that you are not wearing outlandish colors, showing too much skin, or wearing too much jewelery (man or woman).

Unprepared answers:

You know the typical job questions, so have them prepared. Know ahead of time your answers to questions such as, "What you strengths are and weaknesses? Where do you see yourself in ten years? What can you bring to the company that nobody else can? What brought you to this part of your career?" These generalized questions will more than likely be asked of you on the big day.

Unprepared questions:

Just as much as must prepare the answers to give to your interviewer, you must also come prepared with questions to ask of the interviewer about the position and the company. Employers want to see people who think on their toes and are interested in the position as much as possible.

Scribble down a few quick questions that you can ask at the appropriate time of the interview so that you are not trying to think on the spot of what to ask.

Forgetting to do your research:

If you are going on a job interview and know nothing about the company, than you are mistaken. You must learn as much as possible about the position and company prior to your interview. Therefore, you will be able to hold an intelligent and well-informed discussion about the company. You will also be able to ask questions more comprehensively.

Slips of the tongue:

Although this mistake is difficult to control, people often make them. People accidentally say the wrong thing, infer the wrong idea, or blurt out an insult or profanity. Any of these slips of the tongue can throw you out of the interview race. Concentrate on what you say, so that it comes out intelligibly and not incoherently.

Ringing of cell phones:

Make sure your phone is off prior to the interview. Nothing is ruder or more disrespectful to an interviewer than listening to another ring.

Checking the time:

Be careful not to glance at your watch or the wall clock in the middle of your interview. Perhaps you should take off your watch prior to the interview so that you have no chance of accidentally looking at your wrist.

Asking about salary too early:

While it probably one of the determining factors in your decision to work at a company, do not jump the gun in the interview to ask about salary. The interviewer will inevitably tell you what salary and benefits come with the job. There are so many people looking for jobs, so if the company sees you as someone who just wants the money and does not necessarily care about the job, it will work against you in the long run.

Giving your demands:

You are on the job interview and you are unemployed (or are looking to chance jobs). You are no diva. Therefore, you should never make demands at an interview. Even if you have reached a professional status that enables you to make certain demands in the workplace, the interview is never the place for it. They will laugh at you and more than likely never call you again. People want to work with compromising and congenial co-workers; not demanding prima donnas.

Telling about other job offers:

While you may think it appears desirable to speak about other companies offering you a position, it may come across as ridiculous in a job interview. It is important to keep matters separate. They may see you as someone who already has a job. Why would they want to give you their position if you've already got one somewhere else.

Dropping too many names:

Again, this "do not" is a double-edged sword. While contacts are vital in the job market, they can often be used for the wrong purposes. You do not want to be seen as

someone who gets by only on his or her connections.

Tardiness:

Do not be late to your job interview. Rather, arrive around 10-15 minutes early so you can settle in, fill out paperwork, and allow your interviewer to get ready.

Fidgeting:

While you will probably be nervous prior to and during your job interview, try not to fidget. It is distracting and reads nervousness and anxiety. No employer wants a fidgety co-worker in the building.

Lying:

While it is common to exaggerate on resumes, it is downright wrong to lie on them and in your job interview. Honesty is the most venerable quality in a worker and an employer and interviewer will respect your truth and honesty. Do not claim to have worked somewhere if you have not, and so on and so forth.

These basic tips, we hope, will help you along in the job interview season. You may already be a fantastic interviewer, but unfortunately do not always get that golden interview. If this is the case, then you must focus on your resume and references. Regardless, the job interview is key in nailing a position and becoming an employee of the corporate workplace of the U.S.

-S.R.Angalaeswari
Second Yr Ece-A

RISE OF TOMORROW'S PRIME MINISTER

Narendra Modi is positioning himself to be prime minister, what sort of leader will he make? We can tell by understanding him as a man. Look on this piece as his biography. Born on 17th september 1950,Modi is 61.His resume says he has a master's in political science.

"Today, vast section of civil society sees in Narendra Modi the next prime minister of India. I hope he will plant more visible footprints on the international seashore. He has to speak of peace and a durable solution to the Kashmir problem with the rulers of Pakistan"

The most durable role was played by the congress government at the centre. A bogus commission was appointed to whitewash the Godhra tragedy to establish that the attack on the train was not the result of a conspiracy of some evil minded Muslims, but an accidental stove fire. This serious crime by the congress government was fully exposed when a special investigation team appointed by the supreme court made their own independent investigation and reiterated that the burning of the pilgrims was a concerted plan by those who must have known that it will inevitably lead to retaliation and atrocities against the minorities,a finding fortified by recent court judgments.

All that the new prime minister Rajiv Gandhi had to say was a defiant," When a big tree falls the earth must shake". Never did the congress leadership apologies for the atrocities and murders. It is the greatness of Sikh community that they have forgiven the congress.

In a corruption ridden country where the Chief minister of corruption is the congress and its leaders , Narendra Modi shines for his impeccable integrity. He has focused his entire energy on building in Gujarat an able administration and good governance , he has achieved phenomenal devolpment and economic growth, and at the same time bolstered social inclusiveness. Through these he has worked hard to regain

the confidence of the minorities, even as the relentless and pervasive hate campaign against him has continued unabated in the electronic media, among the fashionable intellectuals and civil society activities, who have become the media sweet threats.

- M.Raj Nivash,
SECOND YR ECE-C

"Science without religion is lame, religion without science is blind." – Albert Einstein

ELECTRONIC WASTE

"Electronic waste" may be defined as discarded computers, office electronic equipment, entertainment device electronics, mobile phones, television sets and refrigerators. This definition includes used electronics which are destined for reuse, resale, salvage, recycling, or disposal. Others define the re-usable (working and repairable electronics) and secondary scrap (copper, steel, plastic, etc.) to be "commodities", and reserve the term "waste" for residue or material which is dumped by the buyer rather than recycled, including residue from reuse and recycling operations.

Because loads of surplus electronics are frequently commingled (good, recyclable, and non-recyclable), several public policy advocates apply the term "e-waste" broadly to all surplus electronics. Cathode ray tubes (CRT) are considered one of the hardest types to recycle.

CRTs have relatively high concentration of lead and phosphors (not to be confused with phosphorus), both of which are necessary for the display. The United States Environmental Protection Agency (EPA) includes discarded CRT monitors in its category of "hazardous household waste" but c onsiders CRTs that have been set aside for testing to be commodities if they are not discarded, speculatively accumulated, or left unprotected from weather and other damage.

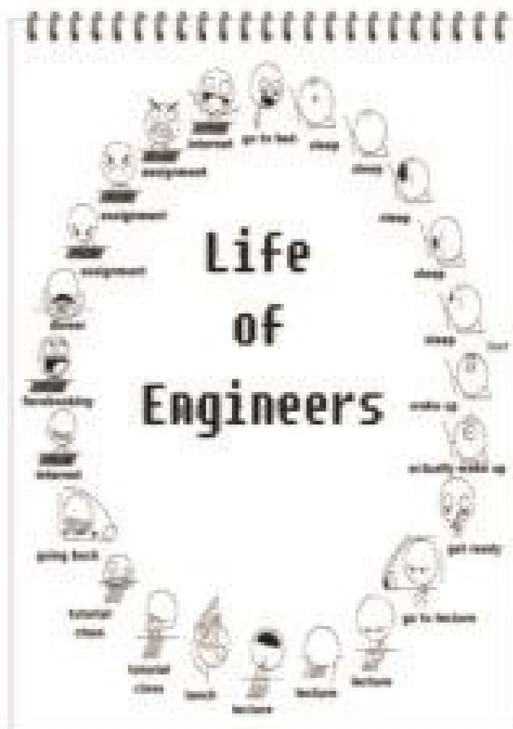
Debate continues over the distinction between "commodity" and "waste" electronics definitions. Some exporters are accused of deliberately leaving difficult-to-recycle, obsolete, or non-repairable equipment mixed in loads of working equipment (though this may also come through ignorance, or to avoid more costly treatment processes). Protectionists may broaden the definition of "waste" electronics in order to protect domestic markets from working secondary equipment.

The high value of the computer recycling subset of electronic waste (working and reusable laptops, desktops, and components like RAM) can help pay the cost of transportation for a larger number of worthless pieces than can be achieved with display devices, which have less (or negative) scrap value. In A 2011 report, "Ghana E-Waste Country Assessment"," found that of 215,000 tons of electronics imported to Ghana, 30% were brand new and 70% were used.

Of the used product, the study concluded that 15% was not reused and was scrapped or discarded. This contrasts with published but uncredited claims that 80% of the imports into Ghana were being burned in primitive conditions

-M.Dhamotharan

Prefinal Yr Ece-A



"An expert is a person who has made all the mistakes that can be made in a very narrow field." – Niels Bohr

POEM'S CORNER

TAMIL POEMS

செய்தித்தாள்

தினமும் NEW வாகப் பிறப்பதினால் என் பெயர்
NEW...S ஓர் உயிரில்லா என்னுள் எத்தனை
உயிருள்ளோரின் உணர்வுகள்!
மலரும் நாளும் ஒன்றோ?
காலையில் பிறந்த மணம் பரப்பி
மாலையிலே உதிர்ந்து போகின்றேன்!
மலரை விரும்பாதோர் உண்டோ?
வெள்ளையனின் மேனியிலே ஓர்
கருப்பனின் நடனம் ! ஆம் !
எனக்குள் நிறக்கொள்கை கிடையாது!
ஏன்? எதற்கு ? எப்படி? என யாரையும்
எதையும் வினவும் எனக்கோ

உயிர் வாழ்க்கை ஒரு நாள் தான் !
இன்று புதியவன் !
நாளை பழையவன்!
பார்த்துப் பார்த்து அச்சடிக்கும் என்னை
மறுநாள் பார்வைக்காகக் கிழிப்பதேன்?
உழவனின் உள்ளதும் உணவுருந்தும்
போதுதான் தெரியும் !

ஆம்!
கருத்துக்களை மக்களின்
மனத்தில் விதைத்து
வளர்க்கும் நானோ
ஓர் ஏழை விவசாயி!
நானோ ! அஞ்சாநெஞ்சன் !

அதிகார வர்க்கம் கூட
என்னைக் கண்டால் சற்று அச்சம் !
விந்தைகளைச் சுமந்து வரும்
வித்தான உலகினை விவரிப்பவன் நான்

அதிசயம் நடப்பின் அடுத்த
அரை நொடியில் பிரசரிப்பேன் !
ஆயினும் என்ன பயன் ?
அன்றே பிறந்து அன்றே அழியும்
ஈசல் போன்றதல்லவா என் வாழ்க்கை!

SR.அங்காள ஈஸ்வரி
இரண்டாமாண்டு 'அ' பிரிவு

குழந்தையின் குழந்தை

அம்மா என்று நான்
சொல்லிய
அத்தருணத்தில்
என்னை பெற்றவள்
அடைந்த மகிழ்ச்சிக்கு
அளவில்லை...

என் முதல் சிரிப்பில்
என் பெற்றோர்
அடைந்த மறுமலர்ச்சி
இன்றும் மலர்கிறது எனது
ஒவ்வொரு வெற்றியிலும்...

அவளின் மகிழ்ச்சியின்
பயணமாய்...
அவள் விரல் பிடித்து
நான்
பதித்த என் பாதசுவடுகள்
என்
மனதில் பதிந்த நினைவு சுவடுகள்
இன்று

அவள் என் விரல் பிடித்து
நடக்கிறாள்
ஒரு குழந்தையாக...
அதே மகிழ்ச்சியுடன்...

த.உ.தனலக்ஷ்மி
மூன்றாமாண்டு 'ஆ' பிரிவு

பெண்

அரும்புமலராய்
அழகாய் மணம் வீசி
அன்பின் ஆழத்தை
அனைவரிடமும் காட்டி
வலம் வருகின்ற
வண்ணத் தேர்!

அகத்தில் அன்புடனும்
புறத்தில் பண்புடனும்
பெண்மைக்கு உண்மையுடனும்
பிரமிக்கும் திறமையுடனும்
பிரகாசிக்கும் சுடர் !

எதையும் இயன்ற வரை
எடுத்து செய்திடும்
உறுதியான உள்ளத்துடன்
உலகை வலம் வரும்
உன்னத படைப்பு !

ரா.முத்துச்செல்வி
இரண்டாமாண்டு 'அ' பிரிவு

"If the facts don't fit the theory, change the facts." — Albert Einstein

அம்மா

கருவறையில்
கடவுளாகத்தான் இருந்தோம்
உன்னை
உதைத்த காரணத்தினாலோ
என்னவோ
சபிக்கப்பட்ட மனிதர்களானோம்...

கடவுள் நம்பிக்கை

இந்தியாவில்
குடிசைகளிலும் குப்பைத்
தொட்டிகளிலும்
லட்சக்கணக்கான "அமுத சம்பந்தர்கள்"
பால் கொடுத்துப் பசியாற்ற
ஒரு பார்வதி தேவிக் கூடயில்லை...

வி.செ.சஹானா
மூன்றாமாண்டு 'அ' பிரிவு

பொறியியலின் புதிய தூதன்

சக்கரம் கண்டறிந்த நாளதனை
அறிவியலின் முதற்படி என்றால்
ரெக்டிஃபையர்கண்டறிந்த பிறெமிங்கொ
இட்டுச் சென்றது - உலகினை
இரண்டாம் படிக்கு !!!

விண்ணோடும் கோள்களிலும் - மனிதன்
உயிர்காக்கும் கருவியிலும் - மக்கள்
உறவாடும் பொறிகளிலும் - குழந்தை
விளையாடும் பொம்மைகளிலும்மென
அங்கிங்கெனாத படி
எங்கும் நிறைந்திருக்கும்
எங்கள் மின்னணுவியலே !!!

ஏனைய பொறியியற் துறைகளை
ஏற்றத்திற்கு அழைத்துச் சென்ற நீ தான்,
பொறியியற் கடவுளின்
புதிய தூதனோ ???

த.விஜயபாரதி
விரிவுரையாளர்

கவிதையின் அழகு

இருளின் நட்சத்திரங்களின் ஒளியில்
கருநிலமென ஒளிரும்
மலையின் நிசப்தங்கள்
பேசுகின்ற வார்த்தைகள்...

நிலவின் நீள் குளிர்ச்சியில்
ஒளி பெய்து போகும்
சப்தங்கள்

குழாயிலிருந்து விழும் நீர்
கை விழுகையில் பேசுவதையும்
உள்ளங்கை வழி வீழ்கையில்
குரல் தொனி மாற்றிக்
கொஞ்சிடும் கொஞ்சல்கள்

எரியும் சுடர்
தொடாத வரை சுட்டு விடாத
வெப்பத்தின் இருப்பு
உச்சரிக்கும் மந்திரங்கள்

அர்த்தமாகி விடும் பொழுதில்
புரிதலுக்குட்பட்ட
அழகிய கவிதையாவேன்...

க.சசிவரன்
மூன்றாமாண்டு 'சி' பிரிவு

இன்றைய இளைஞர்களின் நிலை

தற்கால மனிதன் நாகரீக மாற்றத்தால் நாடோடி வாழ்வை விடுத்து ஓரிடத்திலிருந்து வாழ முற்பட்டான் தற்காலத்தில் மனிதன் காலமாற்றத்திற்கேற்ப தன் நாகரீகத்தை மாற்றிக்கொண்டே செல்கிறான். சமீபகாலமாக இளைஞர்களிடம் நாகரீகம் என்ற பெயரால் ஓர் அவலம் வெகுவாக வேகமாக வளர்ந்து ஒழுக்கத்தையும் வாழ்க்கையையும் ஒரு வழியில் அழித்து கொண்டே செல்கிறது என்று சொன்னால் மிகையாகாது அது தான் மது பழக்கம்.

இந்தியாவின் தூண்கள் என்று கூறப்படும் இளைஞன் தன்னை அனைவருக்கும் முன்பாக தன் நண்பர்களிடம் நாகரீகம் உடையவனாக காட்டிக் கொள்ள பயன்படுத்தும் பழக்கம் தான் மது. இதில் வேதனையான கரு என்னவென்றால் தற்போது உள்ள இளை தலைமுறையினர் 15 முதல் 19 வயது உள்ளோர் அதிகமாக இப்பழக்கத்திற்கு ஆளாகி உள்ளனர் இம்மதுவினால் இந்திய சமூகம் சந்திக்கும் அவலங்களை பார்போம்.

"An eye for an eye will only make the whole world blind." — Mahatma Gandhi

அசாமில் நடந்த அவலம் பற்றி இளைஞர்கள் குடிபோதையில் நடுரோட்டில் இரவில் ஒரு பெண்ணை பாலியல் வன்கொடுமைக்கு தள்ளியது நாட்டையே உலுக்கிய சம்பவம்.
ஒரு பெண் எப்போது சுதந்திரமாக நிறைய நகைகளுடன் இரவு 12 மணிக்கு பாதுகாப்பாக செல்கிறாளோ அன்று தான் உண்மையான சுதந்திரம் என்றார் காந்தியடிகள் ஆனால் இன்று வெறும் சுதந்திரனால் கூட பெண்களால் வெளியில் செல்ல முடியவில்லை.

தலைநகர் டெல்லியில் ஓடும் பேருந்தில் மருத்துவக் கல்லூரி மாணவி ஒரு கும்பலால் பாலியல் பலாத்காரம் செய்து தாக்கி வீசப்பட்டாள். அந்த கும்பலில் ஒரு சிறுவன் வயது 17 மது அருந்தியிருந்தான் என்பது திடுக்கிடும் தகவல் எங்கே சென்று கொண்டிருக்கிறது இளைய சமுதாயம் என்று எண்ணத்தூண்டுகிறது.

தமிழ்நாட்டில் தூத்துக்குடி மாவட்டத்தில் ஒன்பதாவது படிக்கும் மாணவியை பாலியல் பலாத்காரம் செய்து கொலை செய்தான் ஒரு கொடூரன் அதுவும் குடிபோதையில் தான் ஊடகங்கள் மூலம் நமக்கு தெரிந்த இவை மூன்றும் நடப்பதற்கு அடிப்படை காரணமாக இருந்திருப்பது மது தான்.

ஒன்று புரிந்து கொள்ளுங்கள் சாதாரணமாக நல்ல நிலையில் இருக்கும் மனிதனுக்கு பயம் இருக்கும் தவறு செய்ய யோசிப்பான் குற்றம் செய்பவனுக்கு சட்டங்கள் இருப்பதும் அவனுக்கு தெரியும் ஆனாலும் தன்னிலை மாறிய போக்குமதுவுக்கு அடிமையாக இருக்கும் நிலையில் தான் பெரும்பாலான குற்றங்கள் நடக்கின்றன.

இந்த மதுப்பழக்கத்தினால் எத்தனை பிரச்சனைகள் எத்தனை குடும்பங்கள் நடுரோட்டிற்கு தள்ளப்படுகின்றது எத்தனை பெண்கள் குடும்ப பொறுப்பை ஏற்கும் இளைஞன் பொறுப்பே இல்லாமல் மதுவுக்கு அடிமை ஆகின்றான். இதற்கு முழுமையான தீர்வு எந்த ஒரு கல்வியும் ஒரு மாணவனின் பொருளாதாரத்தை உயர்த்துவதற்கு மட்டும் என்ற எண்ணத்தை மாற்றி வாழ்க்கை கல்வியையும் கூட ஒழுக்கத்தையும், கலாச்சாரக்கல்வியையும் தன்னலம் இல்லாமல் பொதுநலத்தையும் தன்னைப்போல் பிறரை நேசி என்ற விதிமுறையில் கல்வி அமைகிறதோ அன்று தான் ஒரு குற்றமற்ற ஒழுக்கமற்ற ஒழுக்கமான இந்தியாவை காணமுடியும்

ச.ராஜா
மூன்றாம் ஆண்டு "ஈ" பிரிவு

TELUGU POEMS

మనసు భావం (Heart Says...)

మనసులోని భావాలెన్నో మరువలేని
గాయాలెన్నో వీడలేని నేస్తాలెన్నో వీడిపోని
బంధాలెన్నో మరపురాని పాటలెన్నో
మధురమయిన క్షణాలెన్నో కవ్వించే
కబుర్లెన్నో మాయమయ్యే మార్పులెన్నో
అవసరానికి ఆడిన అబద్ధాలెన్నో తుంటరిగా
చేసిన చిలిపి పనులెన్నో ఆశ్చర్యపరిచే
అద్భుతాలెన్నో మాటల్లో చెప్పలేని
ముచ్చట్లెన్నో ముసుగు వేసిన మనసుకు
మరుపురాని జ్ఞాపకాలెన్నో ఎన్నో ఎన్నెన్నో
ఇంకెన్నో... మనిషి జీవితంలో మరువలేనివి
ఇంకెన్నో ఇదే జీవితం... దీనిని
అనుభవించు అనుక్షణం

P.S.MADHURI
PRE-FINAL YR ECE-B

స్నేహం (Friend...!)

అందమైనవి అక్షరాలు - అందుకోలేని దూరాలుగా
మరిచిపోలేని జ్ఞాపకాలుగా
మనసుపాడిన ఆటలు మధ్యగా
మనం చేసిన అల్లరి సద్దుగా
నిజంచెబుతున్నా ఆ దేవుడి సాక్షిగా
ఓ... నేస్తం తిరిగిరాదు ఈ సమయం
అందుకే అందుకో ఈ హస్తం
కడదాకా కాదు ... మరుజన్మదాకా కాదు...

V.SAHITHI
PRE-FINAL YR ECE-B

HINDI POEMS

जब भी ये दिल उदास होता है

(feelings of heart)

जब भी ये दिल उदास होता है,
जाने कौन आस-पास होता है ।
होठ चुपचाप बोलते हैं जब,
साँसे कुछ तेज तेज चलती हैं,
आँखे जब दे रही हो आवाजे,
ठंडी आहों में आस जलती है,
आँखों में तरती है तस्वीरें,
तेरा चेहरा तेर खयाल लिए ।
आईना देखता है जब मुझको
एक मासूम सा सवाल लिए,
कोई वादा नहीं किया लेकिन
क्यो तेरा इन्तजार रहता है
बेवजह जब करार मिल जाए,
दिल बड़ा बेकरार रेहता है
जब भी ये दिल उदास होत है,
जाने कौन आस-पास होता है ।

ANISHANAND

SECOND YR ECE-A

कोशिशकरनेवालोंकी

(Hard Work)

लहरोंसेडरकरनौकापारनहींहोती
कोशिशकरनेवालोंकीकभीहारनहींहोती।

नन्हींचींटीजबदानालेकरचलतीहै
चढ़तीदीवारोंपरसौबारफिसलतीहै।
मनकाविश्वासरगोंमेंसाहसभरताहै
चढ़करगिरना,गिरकरचढ़नानअखरताहै।
आखिरउसकीमेहनतबेकारनहींहोती
कोशिशकरनेवालोंकीकभीहारनहींहोती।

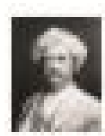
डुबकियांसिंधुमेंगोताखोरलगाताहै
जाजाकरखालीहाथलौटकरआताहै।
मिलतेनसहजहीमोतीगहरेपानीमें
बढ़तादुगनाउत्साहइसीहैरानीमें।
मुट्ठीउसकीखालीहरबारनहींहोती
कोशिशकरनेवालोंकीकभीहारनहींहोती।

असफलताएकचुनौतीहैइसेस्वीकारकरो
क्याकमीरहगई,देखोऔरसुधारकरो।
जबतकनसफलहोनींदचैनकोत्यागोतुम
संघर्षकामैदानछोड़करमतभागोतुम।
कुछकियेबिनाहीजयजयकारनहींहोती
कोशिशकरनेवालोंकीकभीहारनहींहोती।

SHIL.PI.JAISWAL

PRE-FINAL YR ECE-B

"*Good friends, good books, and a sleepy conscience: this is the ideal life.*"– Mark Twain



PENCIL SKETCHES

Prize Winner

R.Eureka Sri
Pre-Final ECE-A

Sriram
Pre-Final Year , ECE-C

Activities in Wireless Mavericks Club from (2010-2013)

Wireless Mavericks Club was inaugurated on 22.4.2010 by the Department of ECE.

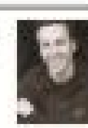
DATE	NAME OF THE EVENT	RESOURCE PERSONS
September 13th 2010	Guest lecture on "**The recent trends in wireless technology**"	Mr.V.Dasarathan, L&T,Mysore
October 22nd 2010	Guest lecture on "**Wireless Networking** "	• Dr.G.Indhumathi,Mepco Schlenk Engg college, • Dr.Tamilselvi,NEC,Kovilpatti
February 2nd week 2011	Guest lecture on "**Wireless Communication**"	• Dr.P.Muthuchidambaranathan, ,NIT,Trichy.
March 9th & 15th 2011	Technical Quiz- Competition	• **Student's Contest**
April 2nd week 2011	Guest lecture on Wireless Networking	Ms.B.Sathyabama, TCE,Madurai.
September 6th 2011	One day seminar on "Wireless Networks"	•Dr.S.J.Thiruvengadam, TCE,Madurai. •Er.A.Ganesan, BSNL,Chennai. •Er.C.V.Vinod,,BSNL,Chennai. Dr.D.Sriramkumar, ,NIT,Trichy.
September 4th week 2011	Guest lecture on Ad hoc Networks	Mr.Ganeshkumar, PSNA college of Engg, Dindugal.
October 3rd week 2011	Guest lecture on "Wireless Networks"	Dr.G.Prema, Mepco Schlenk Engg college, Sivakasi
July 19th 2012	Guest Lecture on "**Wireless ad-hoc Networks &its application**"	•Mr.Narendran Rajagopalan NIT,Trichy.

August 22^{nd} 2012	One day Workshop on **"Networking of Real CISCO devices"**	Mr.P.Athideerapandian , Pandian Systems and Solutions Pvt.Ltd, Maduari.
October 1^{st} 2012	**WII-PRESSENZA&ELECTROCOMM**	Dr.S.Jeyapragasam, Director, IGINP.
February 25^{th} 2013	**KLU-NATURA'13**(Inter department Student fest-An ECO friendly Initiative)	Dr.S.Jeyapragasam, Director, IGINP.
March 13^{th} 2013	Open Quiz Contest-13	**Student's Contest**

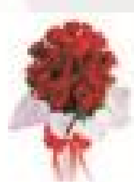
EVENTS ORGANIZED BY IETE CLUB

DATE	NAME OF THE EVENT	RESOURCE PERSONS
August 05^{t} 2009	Inauguration of IETE Student's Forum (ISF)	• Dr. V. Abhai Kumar, Thiyagarajar College of Engineering, Madurai.
February 19^{th} 2010	Technical Aptitude Test	**Student's Contest**
September 13^{th} & 14^{th} 2010	Guest Lecture on "Recent Trends in Wireless Technologies"	• Mr. V. Dasarathan, Larsen & Toubro Ltd., Mysore. • Mrs. Arulmozhi, Saranathan Engineering College, Trichy.
October 27^{th} & 28^{th} 2010	Technical Quiz Preliminary Test Final Quiz	**Student's Contest**
March 23^{rd} 2011	Guest Lecture on "Soft Computing"	• Dr.Hitesh Datt Mathur, Electrical & Electronics Engineer Group, Birla Institute of Technology & Science, Pilani.
April 08^{th} 2011	Guest Lecture on "Design of Embedded Systems"	• Dr. N. Senthil Kumar, Mepco Schlenk Engineering College, Sivakasi.

"Sometimes people are beautiful. Not in looks. Not in what they say. Just in what they are."
— Markus Zusak



September 29th 2011	Quiz Competition	**Student's Contest**
October 15th 2011	Technical Aptitude Test	**Student's Contest**
January 24th & 25th 2012	National conference on VLSI, Embedded and Communication System (NCVEC'12)	•**Dr. Lawrence Jenkins** IISc, Bangalore •**Dr.B.Subramonian** Space Tribology,ISRO. •**Dr. P. G. Krishna Mohan** Professor & Head of the Department, Jawajarlal Nehru TechnologicalUniversity, Kukatpally •**Dr.S.Varada Rajan** Department of Electrical & Electronics Engineering,Sri Venkateswara University, Tirupati.
February 25th 2013	KLU-NATURA'13 Green Campus- An Eco Friendly Initiative	**Student's Contest**
March 13th 2013	Quiz Competition	**Student's Contest**



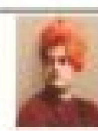
WITH BEST COMPLIMENTS FROM

WIRELESS MAVERICKS CLUB

&

INSTITUTION OF ELECTRONICS AND TELECOMMUNICATION ENGINEERS (IETE)

"Take risks in life: If you win, you can lead! If you loose, you can guide."
- Swami Vivekanando

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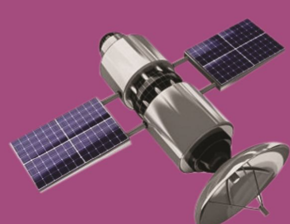

Talent wins games; but team work & intelligence wins championships
- The Magazine Team

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